

Calculus - I (MA1110)

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Preface

These notes have been prepared for the course **Calculus I (MA1110)**, a 1-credit course offered during Segments 1 – 2 of the Fall Semester 2026 at the Indian Institute of Technology Hyderabad (IITH). They are based on the lectures delivered in the course, and their primary purpose is to provide a concise and organized presentation of the topics prescribed in the course syllabus.

These notes are neither self-contained nor intended to serve as a comprehensive textbook on calculus. There are many excellent textbooks on the subject that provide a much broader and more detailed treatment. Students are encouraged to consult these standard references for additional examples, exercises, and deeper insights.

Many of the definitions, examples, exercises, and proofs presented in these notes have been adapted from or inspired by several standard texts, including *Principles of Mathematical Analysis* by Walter Rudin, *Introduction to Real Analysis* by Sadhan Kumar Mapa, and *A Basic Course in Real Analysis* by Ajit Kumar and S. Kumaresan, lecture notes by Prof. Bhakti Bhusan Manna, among others. I gratefully acknowledge these valuable sources.

The material presented here is intended to complement the classroom lectures by emphasizing the fundamental concepts, definitions, theorems, proofs, and illustrative examples discussed during the course. Every effort has been made to keep the exposition clear, concise, and accessible to first-year B.Tech students.

Despite every effort to ensure accuracy, these notes may still contain typographical errors or other inadvertent mistakes. I would be grateful if readers could bring any such errors or suggestions for improvement to my attention by email.

Finally, I would like to acknowledge the assistance of AI tools in preparing the L^AT_EX typesetting and formatting of these notes, which significantly reduced the time required for their preparation.

1 Introduction

Question: What is Calculus?

Answer: Calculus is the Mathematics of change.

We may ask: How fast is a car moving at this instant? How much water has flowed into a tank up to this time? How does the temperature of a room change? How do engineers optimize the shape of a bridge or an aircraft wing? These are all questions involving continuous change. Whenever something changes continuously, calculus provides the mathematical language to study and analyze it.

Where is Calculus Used?

Calculus finds applications in almost every branch of science and engineering. Some examples include:

- Mechanical Engineering: velocity, acceleration, and the motion of machines.
- Electrical Engineering: current and voltage varying with time, and signal processing.
- Computer Science: optimization, machine learning, and neural networks.
- Civil Engineering: beam deflection and structural analysis.
- Chemical Engineering: reaction rates and diffusion.
- Data Science: optimization algorithms and error minimization.

These are only a few examples; calculus is a fundamental tool in many other disciplines as well.

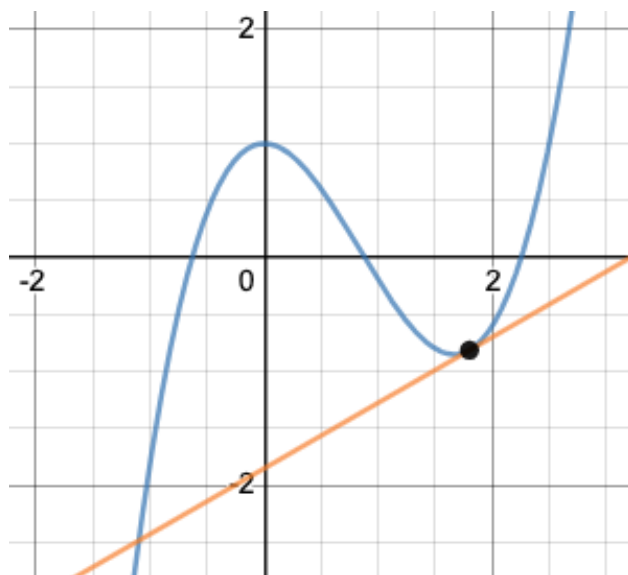
Calculus is a branch of mathematics concerned with the study of sequences, functions, limits, continuity, derivatives, integrals, and infinite series. It provides the mathematical language for understanding and analyzing continuous change.

Sir Isaac Newton (1642–1727), an English mathematician, physicist, and astronomer, and Gottfried Wilhelm Leibniz (1646–1716), a German mathematician, philosopher, and scientist, independently developed the foundations of calculus during the late seventeenth century.

1.1 Syllabus

The syllabus of the course is as follows. The course code is MA1110, and in this course, we will mainly discuss these three topics:

1. Sequence of real numbers: $\{a_1, a_2, \dots\}$.
2. Series of real numbers: $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$.
3. Differential calculus, where we will discuss continuity and differentiability.



1. Sequence of Real Numbers.

By a sequence of real numbers, we mean an enumerated infinite collection of real numbers indexed by the set of natural numbers. Here, we will study the notion of limits for a sequence. If a sequence has a limit, then we call such a sequence a convergent sequence. We will study various properties of convergent sequences, and we will see many examples. Next, we study monotone sequences, that is, either monotone increasing or monotone decreasing sequences. At the end of this section, we give a criterion called the Cauchy criterion, which gives a necessary and sufficient condition for a sequence to be convergent without having any knowledge of the limit.

2. Series of Real Numbers.

In the second section, we discuss a series of real numbers. By a series of real numbers, we roughly mean adding infinitely many real numbers. The study of series is a major part of calculus. Series are used in most areas of mathematics, even for studying finite structures such as combinatorics through generating functions. Infinite series are also widely used in other disciplines such as physics, computer science, statistics, and finance.

For a long time, the idea that such a potentially infinite summation could produce a finite result was considered paradoxical. This paradox was resolved using the concept of a limit during the 17th century. This infinite summation can have a finite value—in this case, the series is called a convergent series; otherwise, it is called a divergent series. We will study various tests to determine whether a series is convergent or divergent.

3. Differential Calculus.

In the last part of this course, we will study differential calculus, which is a subfield of calculus that studies the rates at which quantities change. It is one of the two traditional divisions of calculus, the other being integral calculus, which is the study of the area beneath a curve. The primary objects of study in differential calculus are the derivative of a function and related notions. The derivative of a function at a chosen point describes the rate of change of the function near that point. Geometrically, this is the tangent line at this point of this curve, and the derivative is given by the slope of this tangent line.

Differential calculus and integral calculus are connected by the Fundamental Theorem of Calculus, which states that differentiation is the reverse process to integration. Differentiation has applications to nearly all quantitative disciplines. For example, in physics, the derivative of the momentum of a body with respect to time equals the force applied to the body. Rearranging this statement leads to the famous equation $F = ma$ associated with Newton's second law of motion. In operations research, derivatives determine the most efficient ways to transport materials and design factories. Derivatives are frequently used to find the maxima and minima of a function. Equations involving derivatives are called differential equations and are fundamental in describing natural phenomena. Derivatives and their generalizations appear in many fields of mathematics, such as complex analysis, functional analysis, differential geometry, measure theory, and abstract algebra.

So, in this course, we will discuss mainly continuity and differentiability of a function of a single variable. We will study the rigorous definition of these two notions, and statements of Rolle's theorem, Lagrange's mean value theorem, and their applications.

1.2 Basic properties of the real numbers

Throughout this course, our focus will be on understanding not only how to solve problems, but also why the fundamental results of calculus are true. This requires a careful study of the real number system, which provides the foundation for the notions of limits, continuity, and differentiability.

Does every bounded sequence converge?

$$(-1)^n$$

Does every bounded increasing sequence converge?

Answer: Yes.

We know the statement is true, but mathematics asks **why it is true**. This naturally motivates the study of the real numbers. Before studying sequences and series, we must understand the number system in which they live. The most important property of the real numbers is **completeness**. Today, we begin with the basic properties of the real numbers.

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c \subset \mathbb{C}$$

The students are expected to be familiar with the algebraic operations on the real numbers, namely addition, additive inverse, multiplication, and multiplicative inverse. They may also wish to briefly revisit the basic notions of sets, relations, binary relations, partially ordered sets, and totally ordered sets. In this course, we shall begin with the fact that $(\mathbb{R}, \leq)$ is a totally ordered set. One of the distinguishing features of the real numbers is that any two real numbers can be compared. The first fundamental property of this order relation is the

Law of Trichotomy

The Law of Trichotomy states that for any two real numbers x and y , exactly one of the following statements holds:

1. $x = y$;
2. $x < y$;
3. $y < x$.

1.2.1 Upper and lower bounds

Definition 1. Let $A \subset \mathbb{R}$ be nonempty. A real number α is called an upper bound of A if for each $x \in A$, we have $x \leq \alpha$.

$$\text{Quantifiers:} \quad \forall x \in A (x \leq \alpha).$$

Geometrically, this means that elements of A are to the left of α on the number line.



Figure 1: α is an upper bound of A

A real number α is not an upper bound of A if there exists at least one $x \in A$ such that $x > \alpha$. In terms of quantifiers

$$\exists x \in A (x > \alpha).$$

Geometrically, this means that we can find an element of A to the right of α on the number line.

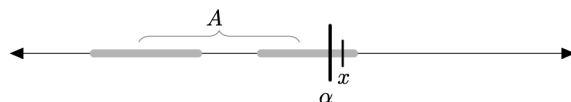


Figure 2: α is not an upper bound of A

If α is an upper bound of A and $\alpha' > \alpha$, then α' is also an upper bound of A .

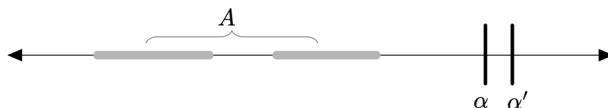


Figure 3: Upper bound of a set is not unique

Exercise 1. Define the notion of a lower bound for a nonempty subset of $\mathbb{R}$.

Exercise 2. Show that $\mathbb{N}$ has a lower bound in $\mathbb{R}$. Does $\mathbb{N}$ have an upper bound in $\mathbb{R}$?

Definition 2. Let $A \subset \mathbb{R}$ be nonempty. We say that A is bounded above in $\mathbb{R}$ if there exists an upper bound of A . Equivalently, there exists $\alpha \in \mathbb{R}$ such that

$$x \leq \alpha, \quad \forall x \in A.$$

In terms of quantifiers, the above definition may be written as

$$\exists \alpha \in \mathbb{R} (\forall x \in A (x \leq \alpha)).$$

Definition 3. Let $A \subset \mathbb{R}$ be nonempty. We say that A is not bounded above in $\mathbb{R}$ if for every $\alpha \in \mathbb{R}$, there exists $x \in A$ (which may depend on α) such that $x > \alpha$.



Figure 4: Set which is not bounded above

In terms of quantifiers, the above definition may be written as

$$\forall \alpha \in \mathbb{R} (\exists x \in A (x > \alpha)).$$

Exercise 3. Using the definition of bounded above, define what it means for a subset of $\mathbb{R}$ to be bounded below. Can you visualize this on the number line? When do you say that a subset $A \subset \mathbb{R}$ is not bounded below in $\mathbb{R}$?

An upper bound of a set need not be an element of the set. For example, if

$$A = (a, b),$$

then b is an upper bound of A , but $b \notin A$. If an upper bound of A is itself an element of A , then it is called the *maximum* of A .

Exercise 4. Suppose that $M, M' \in A$ are both maximum of a nonempty subset $A \subset \mathbb{R}$. Prove that $M = M'$.

Exercise 5. Define the minimum of a nonempty subset $A \subset \mathbb{R}$. Prove that if A has a minimum, then it is unique.

Theorem 1. Let $A \subset \mathbb{R}$ be a nonempty set. If α is a lower bound of A and β is an upper bound of A , then

$$\alpha \leq \beta.$$

Proof. Since A is nonempty, there exists an element $a \in A$. As α is a lower bound of A , we have

$$\alpha \leq a.$$

Similarly, since β is an upper bound of A , we have

$$a \leq \beta.$$

Hence,

$$\alpha \leq a \leq \beta,$$

and therefore

$$\alpha \leq \beta.$$

□

Definition 4. A subset $A \subseteq \mathbb{R}$ is said to be bounded if it is both bounded above and bounded below. Equivalently, there exist real numbers m and M such that $m \leq x \leq M$ for every $x \in A$.

Definition 5. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. A real number $\alpha \in \mathbb{R}$ is called the least upper bound (LUB) or supremum of A if

1. α is an upper bound of A ; and
2. if β is any upper bound of A , then $\alpha \leq \beta$.

Notationally we write: $\alpha = \sup A$ or $\alpha = \text{lub } A$.

Definition 6. Let $A \subset \mathbb{R}$ be a nonempty set that is bounded below. A real number $\alpha \in \mathbb{R}$ is called the greatest lower bound (GLB) or infimum of A if

1. α is a lower bound of A ; and
2. if β is any lower bound of A , then $\beta \leq \alpha$.

Notationally we write: $\alpha = \inf A$ or $\alpha = \text{glb } A$.

Example 1. Let

$$A = (0, 1).$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Notice that neither 0 nor 1 belongs to A .

Example 2. Let

$$A = [0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

In this case,

$$\max A = 1, \quad \min A = 0.$$

Thus, the supremum and infimum belong to the set.

Example 3. *Let*

$$A = (0, 1].$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum since $0 \notin A$.

Example 4. *Let*

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Then

$$\sup A = 1, \quad \inf A = 0.$$

Moreover,

$$\max A = 1,$$

but A has no minimum.

Example 5. *Let*

$$A = \mathbb{N}.$$

Then A is bounded below but not bounded above. Here, $\inf A = 1$, whereas $\sup A$ does not exist in $\mathbb{R}$.

Example 6. *Let*

$$A = \{x \in \mathbb{R} : x^2 < 4\}.$$

Since

$$A = (-2, 2),$$

we have

$$\sup A = 2, \quad \inf A = -2.$$

However, A has neither a maximum nor a minimum.

1.2.2 LUB property of the real numbers

Let $A \subset \mathbb{R}$ be nonempty and bounded above. Then there exists $\alpha \in \mathbb{R}$ such that

$$\alpha = \sup A \text{ (or lub } A).$$

$\mathbb{Q}$ does not satisfy the Least Upper Bound Property.

Exercise 6. Construct a nonempty subset of $\mathbb{Q}$ that is bounded above but whose least upper bound does not belong to $\mathbb{Q}$.

Answer: Take $A = \{x \in \mathbb{Q} : x^2 < 2\}$, then $\text{lub } A = \sqrt{2} \notin \mathbb{Q}$.

This example illustrates that the rational numbers have **gaps**, whereas the real numbers do not.

The above result is known as the **Completeness Property** (or **Least Upper Bound Property**) of the real numbers. It is one of the fundamental properties that distinguishes the real number system from the rational number system. In this course, we shall accept this property without proof and use it throughout our study of calculus. A proof of this result requires a rigorous construction of the real numbers, for example, using Dedekind cuts or equivalence classes of Cauchy sequences of rational numbers, which we shall not discuss in this course.

Theorem 2. The set $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

Proof. Suppose, for contradiction, that $\mathbb{N}$ is bounded above in $\mathbb{R}$. Then, by the Completeness Property, the set $\mathbb{N}$ has a least upper bound, say α .

Since $\alpha - 1$ is not an upper bound of $\mathbb{N}$, there exists $n \in \mathbb{N}$ such that

$$n > \alpha - 1.$$

Hence,

$$n + 1 > \alpha.$$

As $n + 1 \in \mathbb{N}$, this contradicts the fact that α is an upper bound of $\mathbb{N}$.

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 3 (Archimedean Property). For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

Proof. Proof by contradiction. Suppose that

$$nx \leq y, \quad \forall n \in \mathbb{N}.$$

Then

$$n \leq \frac{y}{x}, \quad \forall n \in \mathbb{N},$$

since $x > 0$. Thus $\frac{y}{x}$ is an upper bound of $\mathbb{N}$, contradicting the previous theorem. Hence there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

□

Exercise 7. Prove that the set of integers $\mathbb{Z}$ is neither bounded above nor bounded below in $\mathbb{R}$.

Recall. For every $x \in \mathbb{R}$, there exists a unique integer $\lfloor x \rfloor$ such that

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

The integer $\lfloor x \rfloor$ is called the *floor* of x . Proving this result using the Archimedean Property is a fun exercise.

Theorem 4 (Density of $\mathbb{Q}$ in $\mathbb{R}$). Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists a rational number $r \in \mathbb{Q}$ satisfying

$$a < r < b.$$

Proof. Since $b - a > 0$, the Archimedean Property guarantees the existence of $n \in \mathbb{N}$ such that

$$n(b - a) > 1.$$

Let

$$k = \lfloor na \rfloor, \quad m = k + 1.$$

Then

$$k \leq na < k + 1 = m,$$

and hence

$$na < m.$$

We claim that $m < nb$. Suppose, on the contrary, that $m \geq nb$. Then

$$1 = (k + 1) - k = m - k \geq nb - na = n(b - a) > 1,$$

which is impossible. Therefore,

$$m < nb.$$

Consequently,

$$na < m < nb.$$

Dividing throughout by the positive integer n , we obtain

$$a < \frac{m}{n} < b.$$

Since $m, n \in \mathbb{Z}$ and $n > 0$, the number

$$r = \frac{m}{n}$$

is rational. This completes the proof. $\square$

Corollary 1. *Let $a, b \in \mathbb{R}$ be such that $a < b$. Then there exists an irrational number $t \in \mathbb{R} \setminus \mathbb{Q}$ satisfying*

$$a < t < b.$$

Proof. Since

$$a - \sqrt{2} < b - \sqrt{2},$$

the density of $\mathbb{Q}$ in $\mathbb{R}$ guarantees the existence of a rational number r such that

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Adding $\sqrt{2}$ throughout, we obtain

$$a < r + \sqrt{2} < b.$$

We claim that $r + \sqrt{2}$ is irrational. Indeed, if $r + \sqrt{2}$ were rational, then

$$\sqrt{2} = (r + \sqrt{2}) - r$$

would be the difference of two rational numbers, and hence rational, a contradiction since $\sqrt{2}$ is irrational. Therefore,

$$t = r + \sqrt{2}$$

is an irrational number satisfying

$$a < t < b.$$

This completes the proof. $\square$

Absolute Value. For $x \in \mathbb{R}$, the *absolute value* of x is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

We now recall some useful properties of the absolute value. For all $a, b \in \mathbb{R}$,

1. $|a| \geq 0$ and $|a| = 0$ iff $a = 0$. Moreover, $-|a| \leq a \leq |a|$.
2. Let $\varepsilon > 0$. Then $|a - b| < \varepsilon$ iff $a \in (b - \varepsilon, b + \varepsilon)$.
3. $|ab| = |a||b|$ and $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, provided $b \neq 0$.
4. Triangle Inequality: $|a + b| \leq |a| + |b|$.
5. Reverse Triangle Inequality: $||a| - |b|| \leq |a - b|$.
6. $\max\{a, b\} = \frac{a+b+|a-b|}{2}$ and $\min\{a, b\} = \frac{a+b-|a-b|}{2}$.

2 Sequences

Sequences arise naturally whenever we wish to approximate a quantity. For example, consider the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

If we successively retain one more decimal place, we obtain the sequence

$$0.1, 0.14, 0.142, 0.1428, 0.14285, \dots$$

Each term is obtained by keeping one additional decimal place of the decimal expansion of $\frac{1}{7}$, and hence provides a better approximation to $\frac{1}{7}$ than the previous one. For instance, if we require the approximation to be accurate up to 10^{-4} , then any term with at least four decimal places is sufficient.

The notion of a sequence and its convergence provides a rigorous mathematical framework for studying such approximations. Intuitively, a sequence (x_n) converges to a real number x if its terms become arbitrarily close to x as n becomes sufficiently large. In other words, given any prescribed level of accuracy, all terms of the sequence beyond a certain stage approximate x to that accuracy.

2.1 Sequences and their convergence

Definition 7. A sequence of real numbers is a function

$$f : \mathbb{N} \rightarrow \mathbb{R}.$$

For each $n \in \mathbb{N}$, we write

$$x_n = f(n),$$

and call x_n the n -th term of the sequence. The sequence is usually denoted by (x_n) or $\{x_n\}$, and may also be written as

$$(x_n)_n, \quad \{x_n\}_n, \quad \text{or} \quad (x_1, x_2, \dots).$$

Example 7. Some examples of sequences are given below.

1. Fix $c \in \mathbb{R}$ and define $x_n = c$ for all $n \in \mathbb{N}$. Then $(x_n) = (c, c, c, \dots)$. This is called a constant sequence.
2. Let $x_n = \frac{1}{n}$. Then $(x_n) = (1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots)$.
3. Let $x_n = \frac{(-1)^{n+1}}{n}$. Then $(x_n) = (1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, \frac{1}{2k-1}, -\frac{1}{2k}, \dots)$.
4. Let $x_n = (-1)^n$. Then $(x_n) = (-1, 1, -1, 1, \dots)$.
5. Let $x_n = n$. Then $(x_n) = (1, 2, 3, 4, \dots)$.
6. Let $x_n = 2^n$. Then $(x_n) = (2, 4, 8, 16, \dots)$.
7. Let $x_n = \frac{1}{2^n}$. Then $(x_n) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots)$.
8. Let $x_1 = x_2 = 1$ and define $x_n = x_{n-1} + x_{n-2}$ for all $n \geq 3$. Then $(x_n) = (1, 1, 2, 3, 5, 8, 13, 21, 34, \dots)$.

Example 8. Let

$$x_1 = 0.1, \quad x_2 = 0.14, \quad x_3 = 0.142, \quad x_4 = 0.1428, \quad \dots,$$

where x_n is obtained by retaining the first n digits after the decimal point in the decimal expansion of

$$\frac{1}{7} = 0.142857142857 \dots$$

Observe that

$$\left| \frac{1}{7} - x_n \right| < \frac{1}{10^n}, \quad \forall n \in \mathbb{N}.$$

Thus, given any prescribed level of accuracy, we can choose n sufficiently large so that x_n approximates $\frac{1}{7}$ to that accuracy. This example motivates the notion of convergence, which we study next.

Definition 8. Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. We say that (x_n) converges to x if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

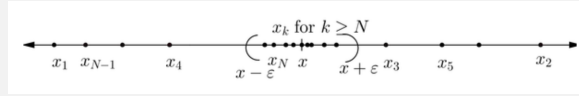
$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

In this case, x is called the limit of the sequence (x_n) . We write

$$x_n \rightarrow x \quad \text{as } n \rightarrow \infty,$$

or equivalently,

$$\lim_{n \rightarrow \infty} x_n = x.$$



In terms of quantifiers: (x_n) converges to x if

$$(\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Remark 1. 1. In the definition of convergence, the positive number ε may be chosen arbitrarily small. Nevertheless, for every $\varepsilon > 0$, there always exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

2. Let $\varepsilon > 0$. If (x_n) converges to x , then there exists $N \in \mathbb{N}$ such that

$$x_n \in (x - \varepsilon, x + \varepsilon), \quad \forall n \geq N.$$

Equivalently, the interval $(x - \varepsilon, x + \varepsilon)$ contains all but finitely many terms of the sequence (x_n) , i.e., only finitely many terms of the sequence can stay outside the interval.

Example 9. Consider the constant sequence $x_n = c$ for all $n \in \mathbb{N}$. We claim that

$$\lim_{n \rightarrow \infty} x_n = c.$$

Indeed, let $\varepsilon > 0$ be given. Since

$$|x_n - c| = 0 < \varepsilon, \quad \forall n \in \mathbb{N},$$

we may simply choose $N = 1$. Hence,

$$|x_n - c| < \varepsilon, \quad \forall n \geq N.$$

Therefore, (x_n) converges to c .

Example 10. Consider the sequence $(\frac{1}{n})$. We claim that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Let $\varepsilon > 0$ be given. We have

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\varepsilon}.$$

Hence, for every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Exercise 8. Using the previous example, show that if $a, b \in \mathbb{R}$ with $a \neq 0$, then the sequence

$$x_n = \frac{1}{an + b}$$

converges. Determine its limit.

Example 11. Consider the sequence

$$a_n = \frac{n^2 + 1}{n^2}.$$

We claim that

$$\lim_{n \rightarrow \infty} a_n = 1.$$

Let $\varepsilon > 0$ be given. We have

$$|a_n - 1| = \left| \frac{n^2 + 1}{n^2} - 1 \right| = \frac{1}{n^2}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ such that

$$N > \frac{1}{\sqrt{\varepsilon}}.$$

Hence, for every $n \geq N$,

$$|a_n - 1| = \frac{1}{n^2} \leq \frac{1}{N^2} < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{n^2 + 1}{n^2} = 1.$$

From the examples above, we see that proving the convergence of a sequence usually has two steps. First, we guess its limit. Then we show that the terms of the sequence get closer and closer to this value by estimating $|x_n - x|$. This raises two natural questions:

How do we guess the limit? & Can a sequence have more than one limit?

The first question becomes easier with practice. We now answer the second question.

Theorem 5 (Uniqueness of the Limit). *If*

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{and} \quad \lim_{n \rightarrow \infty} x_n = y,$$

then $x = y$.

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

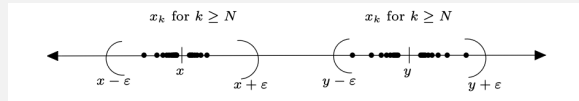
Similarly, since $x_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$, the triangle inequality gives

$$|x - y| = |x - x_n + x_n - y| \leq |x - x_n| + |x_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $x = y$.



□

Example 12. Consider the sequence $x_n = n$. We claim that this sequence does not converge.

Assume, on the contrary, that (x_n) converges to some $x \in \mathbb{R}$. By Remark 1, following the definition of convergence, the interval

$$(x - 1, x + 1)$$

must contain all but finitely many terms of the sequence. Hence, there exists $N \in \mathbb{N}$ such that

$$n = x_n \in (x - 1, x + 1), \quad \forall n \geq N.$$

In particular,

$$n < x + 1, \quad \forall n \geq N.$$

Thus, the set $\mathbb{N}$ is bounded above by $\max\{N, x + 1\}$, contradicting the fact that $\mathbb{N}$ is not bounded above. Therefore, (x_n) does not converge.

Example 13. Consider the sequence $x_n = (-1)^n$. Observe that the intervals

$$\left(-\frac{3}{2}, -\frac{1}{2}\right) \quad \text{and} \quad \left(\frac{1}{2}, \frac{3}{2}\right)$$

each contain infinitely many terms of the sequence. Hence, no interval containing a possible limit can contain all but finitely many terms of the sequence. Therefore, $((-1)^n)$ is not convergent.

Converge

$$\lim_{n \rightarrow \infty} x_n = x \text{ if } (\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Diverge

$$(\forall x \in \mathbb{R}) (\exists \varepsilon > 0) (\forall N \in \mathbb{N}) (\exists n \geq N) (|x_n - x| \geq \varepsilon).$$

A sequence that is not convergent is called *divergent*. This does *not* mean that the terms of the sequence must tend to infinity. A divergent sequence may oscillate, as in the sequence $((-1)^n)$, or it may become arbitrarily large, as in the sequence (n) . Thus, there are different ways in which a sequence can fail to converge. To prove that a sequence is convergent, we must identify its limit and verify the definition of convergence. On the other hand, to prove that a sequence is divergent, it is enough to show that no real number can be its limit.

2.2 Properties of convergent sequences

Definition 9. A sequence (x_n) of real numbers is said to be bounded if there exists a constant $C > 0$ such that

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Example 14. The following are some examples of bounded and unbounded sequences.

1. Every constant sequence is bounded.
2. The sequence $\left(\frac{1}{n}\right)$ is bounded.
3. The sequence $((-1)^n)$ is bounded.
4. The sequence $\left(\frac{n^2+1}{n^2}\right)$ is bounded.
5. The sequence $(\sin n)$ is bounded.

Definition 10. A sequence (x_n) of real numbers is said to be unbounded if, for every $C > 0$, there exists $n \in \mathbb{N}$ such that $|x_n| > C$.

Example 15. The sequences

$$(n^2), \quad (n^n), \quad (n)$$

are unbounded.

Theorem 6. Every convergent sequence of real numbers is bounded.

Proof. Let (x_n) be a sequence converging to $x \in \mathbb{R}$. Choose $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < 1, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x_n| = |x_n - x + x| \leq |x_n - x| + |x| < 1 + |x|, \quad \forall n \geq N.$$

Now let

$$C = \max\{|x_1|, \dots, |x_{N-1}|, 1 + |x|\}.$$

Then

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Hence, (x_n) is bounded. □

The converse of the above result is not true. Can you recall an example?

Theorem 7. Let (x_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} x_n = x,$$

where $x \neq 0$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

Proof. Choose

$$\varepsilon = \frac{|x|}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{|x|}{2}, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x| = |x - x_n + x_n| \leq |x - x_n| + |x_n| < \frac{|x|}{2} + |x_n|.$$

Hence,

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

This completes the proof. □

Theorem 8. Let (a_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} a_n = a.$$

If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

Proof. Suppose, on the contrary, that $a < 0$. Then there exists $\delta > 0$ such that

$$a + \delta < 0.$$

Since $a_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that

$$a_n \in (a - \delta, a + \delta), \quad \forall n \geq N.$$

Since $a + \delta < 0$, every point in the interval $(a - \delta, a + \delta)$ is negative. Thus,

$$a_n < 0, \quad \forall n \geq N,$$

which contradicts the assumption that $a_n \geq 0$ for all $n \in \mathbb{N}$. Therefore, $a \geq 0$. □

Exercise 9. Let (u_n) and (v_n) be two convergent sequences such that

$$u_n \leq v_n, \quad \forall n \in \mathbb{N}.$$

Show that

$$\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n.$$

Given two sequences (x_n) and (y_n) , we can construct new sequences by adding, multiplying, or multiplying by a real number. A natural question is: if the original sequences are convergent, what can we say about these newly constructed sequences? In this section, we show that convergence is preserved under these basic algebraic operations.

Theorem 9 (Algebra of Convergent Sequences). Let (x_n) and (y_n) be sequences of real numbers such that $\lim_{n \rightarrow \infty} x_n = x$, $\lim_{n \rightarrow \infty} y_n = y$, and let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$.
2. $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x$.
3. $\lim_{n \rightarrow \infty} (x_n y_n) = xy$.
4. If $x_n > 0$ for all $n \in \mathbb{N}$ and $x > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}$.

Proof. Let $\varepsilon > 0$ be given.

(1) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

Similarly, since $y_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|y_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y.$$

(2) If $\alpha = 0$, then the result is obvious. Assume $\alpha \neq 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{|\alpha|}, \quad \forall n \geq N.$$

Hence,

$$|\alpha x_n - \alpha x| = |\alpha| |x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x.$$

(3) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n| - |x| \leq ||x_n| - |x|| \leq |x_n - x| < 1, \quad \forall n \geq N_1.$$

Also, there exist $N_2, N_3 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2(1 + |y|)}$$

for all $n \geq N_2$, and

$$|y_n - y| < \frac{\varepsilon}{2(1 + |x|)}$$

for all $n \geq N_3$. Let $N = \max\{N_1, N_2, N_3\}$. Then

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x_n y + x_n y - xy| \\ &\leq |x_n| |y_n - y| + |y| |x_n - x| \\ &\leq (|x| + 1) \frac{\varepsilon}{2(1 + |x|)} + |y| \frac{\varepsilon}{2(1 + |y|)} < \varepsilon. \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n y_n) = xy.$$

(4) Since $x > 0$, by the previous Theorem 7 there exists $N_1 \in \mathbb{N}$ such that

$$x_n > \frac{x}{2}, \quad \forall n \geq N_1.$$

Also, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{x^2 \varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\left| \frac{1}{x_n} - \frac{1}{x} \right| = \frac{|x_n - x|}{x x_n} \leq \frac{2}{x^2} |x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}.$$

□

Exercise 10. In Part (4) of the above theorem, the assumption $x > 0$ is essential. What can go wrong if $x = 0$? Give an example to show that the conclusion need not hold.

2.3 Cauchy sequences

So far, every time we proved that a sequence converges, we first guessed its limit and then verified the definition of convergence. This naturally raises the following question:

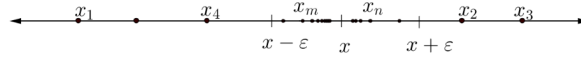
Can we decide whether a sequence is convergent without knowing its limit?

Suppose that (x_n) converges to x . Let $\varepsilon > 0$ be given. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Hence, for any $m, n \geq N$, the triangle inequality gives

$$|x_n - x_m| \leq |x_n - x| + |x_m - x| < \varepsilon + \varepsilon = 2\varepsilon.$$



Thus, when the indices are sufficiently large, the terms of a convergent sequence become arbitrarily close to each other. This observation motivates the following definition.

Definition 11 (Cauchy Sequence). A sequence (x_n) of real numbers is said to be Cauchy if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon, \quad \forall m, n \geq N.$$

Example 16. Every convergent sequence is a Cauchy sequence. Is the converse true?

Theorem 10. [Cauchy Criterion] A sequence of real numbers is Cauchy if and only if it is convergent.

Proof. We have already seen that every convergent sequence is Cauchy. We now prove the converse.

Let $E = \{x \in \mathbb{R} : \exists N \in \mathbb{N} \text{ such that } x < x_n, \forall n \geq N\}$.

Let $\delta > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_m - x_n| < \delta$ for all

$m, n \geq N_0$. In particular, $|x_{N_0} - x_n| < \delta$ for all $n \geq N_0$. Hence, $x_n \in (x_{N_0} - \delta, x_{N_0} + \delta)$ for all $n \geq N_0$. In particular, $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Claim 1. $x_{N_0} - \delta \in E$.

Indeed, taking $N = N_0$, we have $x_n > x_{N_0} - \delta$ for all $n \geq N_0$.

Claim 2. $x_{N_0} + \delta$ is an upper bound of E .

Suppose not. Then there exists $x \in E$ such that $x > x_{N_0} + \delta$. Since $x \in E$, there exists $N \in \mathbb{N}$ such that $x_n > x$ for all $n \geq N$. Hence, for all $n \geq \max\{N, N_0\}$,

$$x_n > x > x_{N_0} + \delta,$$

which contradicts the inequality $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Thus, E is nonempty and bounded above. By the completeness of $\mathbb{R}$, let $\ell = \sup E$.

Claim 3. $x_n \rightarrow \ell$.

Let $\varepsilon > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_n - x_{N_0}| < \varepsilon/2$ for all $n \geq N_0$.

By Claim 1, $x_{N_0} - \varepsilon/2 \in E$, and hence $x_{N_0} - \varepsilon/2 \leq \ell$. By Claim 2, $\ell \leq x_{N_0} + \varepsilon/2$. Therefore, $|x_{N_0} - \ell| \leq \varepsilon/2$.

Now, for every $n \geq N_0$,

$$|x_n - \ell| \leq |x_n - x_{N_0}| + |x_{N_0} - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $x_n \rightarrow \ell$. □

Example 17. The sequence $(\frac{1}{n})$ is a Cauchy sequence.

Proof. Let $\varepsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then, for any $m, n \geq N$, without loss of generality, assume $m \leq n$. Hence,

$$\left| \frac{1}{m} - \frac{1}{n} \right| = \frac{1}{m} - \frac{1}{n} < \frac{1}{m} \leq \frac{1}{N} < \varepsilon.$$

Therefore, $(\frac{1}{n})$ is a Cauchy sequence. □

Example 18. Let

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \quad n \in \mathbb{N}.$$

We show, using the Cauchy Criterion, that (s_n) is not convergent.

Proof. By the Cauchy Criterion, it is enough to show that (s_n) is not a Cauchy sequence. Choose $\varepsilon = \frac{1}{2}$ and let $N \in \mathbb{N}$ be arbitrary. Take $n \geq N$ and let $m = 2n$.

Then

$$\begin{aligned} |s_m - s_n| &= |s_{2n} - s_n| = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \\ &\geq \underbrace{\frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ terms}} = \frac{1}{2}. \end{aligned}$$

Thus, for every $N \in \mathbb{N}$, there exist $m, n \geq N$ such that

$$|s_m - s_n| \geq \frac{1}{2}.$$

Hence, (s_n) is not a Cauchy sequence. Therefore, by the Cauchy Criterion, (s_n) is not convergent. $\square$

Example 19. *The sequence $((-1)^n)$ is not a Cauchy sequence.*

Proof. Choose $\varepsilon = 1$. Let $N \in \mathbb{N}$ be arbitrary. Take

$$m = \begin{cases} N+1, & \text{if } N \text{ is even,} \\ N+2, & \text{if } N \text{ is odd,} \end{cases} \quad \text{and} \quad n = m+1.$$

Then $m, n \geq N$, $(-1)^m = 1$, and $(-1)^n = -1$. Hence,

$$|(-1)^m - (-1)^n| = |1 - (-1)| = 2 > 1 = \varepsilon.$$

Therefore, $((-1)^n)$ is not a Cauchy sequence. $\square$

2.4 Monotone sequences

Definition 12. *A sequence (x_n) is said to be increasing if*

$$x_n \leq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 13. *A sequence (x_n) is said to be strictly increasing if*

$$x_n < x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 14. *A sequence (x_n) is said to be decreasing if*

$$x_n \geq x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 15. A sequence (x_n) is said to be strictly decreasing if

$$x_n > x_{n+1}, \quad \forall n \in \mathbb{N}.$$

Definition 16. A sequence is said to be monotone if it is either increasing or decreasing.

Remark 2. Every increasing sequence is bounded below by x_1 . Hence, an increasing sequence is bounded if and only if it is bounded above. Similarly, every decreasing sequence is bounded above by x_1 . Therefore, a decreasing sequence is bounded if and only if it is bounded below.

Example 20. The following are examples of monotone sequences.

1. The constant sequence (c) is both increasing and decreasing.
2. $(\frac{1}{n})$ is strictly decreasing.
3. (n) is strictly increasing.
4. (2^n) is strictly increasing.
5. $(\frac{n^2+1}{n^2}) = (1 + \frac{1}{n^2})$ is strictly decreasing.
6. $((-1)^n)$ is not monotone.

Theorem 11 (Monotone Convergence Theorem). An increasing sequence of real numbers is convergent if and only if it is bounded above.

Proof. Let (x_n) be an increasing sequence that is bounded above. We show that (x_n) is convergent.

Let

$$E = \{x_n : n \in \mathbb{N}\}.$$

Since E is nonempty and bounded above, by the completeness property of $\mathbb{R}$, the set E has a least upper bound. Let

$$\ell = \sup E.$$

Let $\varepsilon > 0$ be given. Since $\ell - \varepsilon$ is not an upper bound of E , there exists $N \in \mathbb{N}$ such that

$$x_N > \ell - \varepsilon.$$

As (x_n) is increasing, for every $n \geq N$,

$$\ell - \varepsilon < x_N \leq x_n \leq \ell < \ell + \varepsilon.$$

Hence,

$$|x_n - \ell| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} x_n = \ell.$$

The converse follows immediately from the fact that every convergent sequence is bounded. $\square$

Exercise 11. Show that a decreasing sequence of real numbers is convergent if and only if it is bounded below.

Notation. The notation

$$x_n \downarrow x$$

means that the sequence (x_n) is decreasing and converges to x . Similarly, the notation

$$x_n \uparrow x$$

means that the sequence (x_n) is increasing and converges to x .

Example 21. Let $0 \leq r < 1$. We show that $\lim_{n \rightarrow \infty} r^n = 0$.

If $r = 0$, then there is nothing to prove. So assume that $0 < r < 1$. Since $r^{n+1} < r^n$ for every $n \in \mathbb{N}$, the sequence (r^n) is decreasing. Moreover, it is bounded below by 0. Hence, by the Monotone Convergence Theorem, (r^n) converges. Let $\lim_{n \rightarrow \infty} r^n = \ell$.

By the Algebra of Convergent Sequences, $\lim_{n \rightarrow \infty} r^{n+1} = r\ell$. On the other hand, r^{n+1} is just the shifted sequence of (r^n) , so $\lim_{n \rightarrow \infty} r^{n+1} = \ell$. By the Uniqueness of Limits, $r\ell = \ell$. Since $0 \leq r < 1$, we have $(1 - r)\ell = 0$, and therefore $\ell = 0$. Hence,

$$\lim_{n \rightarrow \infty} r^n = 0.$$

Example 22. Let $0 \leq r < 1$, and define $s_n := 1 + r + r^2 + \cdots + r^n$.

If $r = 0$, then (s_n) is the constant sequence 1, and hence it is convergent. So assume that $0 < r < 1$. Since $s_{n+1} = s_n + r^{n+1} > s_n$, the sequence (s_n) is increasing.

Moreover, $s_n = \frac{1 - r^{n+1}}{1 - r} \leq \frac{1}{1 - r}$, so (s_n) is bounded above. Hence, by the Monotone Convergence Theorem, (s_n) converges.

Finally, $s_n = \frac{1 - r^{n+1}}{1 - r} = \frac{1}{1 - r} - \frac{r^{n+1}}{1 - r}$. Since $\lim_{n \rightarrow \infty} r^{n+1} = 0$, it follows from the Algebra of Convergent Sequences that

$$\lim_{n \rightarrow \infty} s_n = \frac{1}{1 - r}.$$

2.5 Sandwich theorem

In many situations, we may be able to guess the limit of a sequence but find it difficult to prove the convergence directly. A useful idea is to compare the given sequence with two simpler sequences whose limits are already known. If the given sequence always lies between these two sequences, then it is forced to have the same limit. This idea is known as the *Sandwich Theorem*.

Theorem 12 (Sandwich Theorem). *Let (x_n) , (y_n) and (z_n) be sequences of real numbers such that*

1. $\lim_{n \rightarrow \infty} x_n = \alpha$ and $\lim_{n \rightarrow \infty} y_n = \alpha$;
2. $x_n \leq z_n \leq y_n$ for all $n \in \mathbb{N}$.

Then

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow \alpha$, there exists $N_1 \in \mathbb{N}$ such that $x_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ such that $y_n \in (\alpha - \varepsilon, \alpha + \varepsilon)$ for all $n \geq N_2$.

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\alpha - \varepsilon < x_n \leq z_n \leq y_n < \alpha + \varepsilon.$$

Hence,

$$|z_n - \alpha| < \varepsilon, \quad \forall n \geq N.$$

Therefore,

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

□

Example 23. *Define the sequence (z_n) by*

$$z_n = \frac{\sin n}{n}, \quad n \in \mathbb{N}.$$

We show that $\lim_{n \rightarrow \infty} z_n = 0$.

Since

$$-1 \leq \sin n \leq 1,$$

we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

We already know that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \left(-\frac{1}{n} \right) = 0.$$

Hence, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$$

Example 24. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \cdots + \frac{1}{\sqrt{n^2 + n}} \right) = 1.$$

Proof. Let $a_n := \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$. Since $\frac{1}{\sqrt{n^2 + n}} \leq \frac{1}{\sqrt{n^2 + k}} \leq \frac{1}{\sqrt{n^2 + 1}}$, for every $1 \leq k \leq n$, we obtain $u_n \leq a_n \leq v_n$, where $u_n := \frac{n}{\sqrt{n^2 + n}}$ and $v_n := \frac{n}{\sqrt{n^2 + 1}}$.

Now, $u_n = \frac{1}{\sqrt{1 + \frac{1}{n}}} \rightarrow 1$, and $v_n = \frac{1}{\sqrt{1 + \frac{1}{n^2}}} \rightarrow 1$. Hence, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} a_n = 1.$$

□

Example 25. Let $a > 0$. We show that

$$\lim_{n \rightarrow \infty} a^{1/n} = 1.$$

Proof. If $a = 1$, there is nothing to prove. Assume first that $a > 1$. Write $a^{1/n} = 1 + x_n$, where $x_n > 0$. Then $a = (1 + x_n)^n$. By Bernoulli's inequality, $(1 + x_n)^n \geq 1 + nx_n$. Hence, $a \geq 1 + nx_n$, which implies $0 < x_n \leq \frac{a - 1}{n}$.

Let $\varepsilon > 0$ be given. By the Archimedean Property, choose $N \in \mathbb{N}$ such that $\frac{a - 1}{N} < \varepsilon$.

Then, for every $n \geq N$, $0 < x_n \leq \frac{a - 1}{n} \leq \frac{a - 1}{N} < \varepsilon$. Therefore, $|a^{1/n} - 1| = x_n < \varepsilon$, and hence $a^{1/n} \rightarrow 1$.

Now suppose that $0 < a < 1$. Then $\frac{1}{a} > 1$, and by the first part, $\left(\frac{1}{a} \right)^{1/n} \rightarrow 1$. Since $a^{1/n} = \frac{1}{(1/a)^{1/n}}$, the Algebra of Convergent Sequences gives $a^{1/n} \rightarrow 1$. □

Example 26. Let $0 < a < b$. We show that

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

Proof. Since $a^n > 0$, we have $b^n \leq a^n + b^n \leq 2b^n$. Taking the n -th root, we obtain $b \leq (a^n + b^n)^{1/n} \leq 2^{1/n}b$.

We already know that $\lim_{n \rightarrow \infty} 2^{1/n} = 1$. Hence, $\lim_{n \rightarrow \infty} 2^{1/n}b = b$. Therefore, by the Sandwich Theorem,

$$\lim_{n \rightarrow \infty} (a^n + b^n)^{1/n} = b.$$

□

Exercise 12. Let (a_n) be a bounded sequence of real numbers and let (x_n) be a sequence such that $x_n \rightarrow 0$.

1. Show that $\lim_{n \rightarrow \infty} a_n x_n = 0$.

2. Deduce that

$$\lim_{n \rightarrow \infty} r^n = 0, \quad \text{for every } |r| < 1.$$

(Hint. Write $r^n = |r|^n \left(\frac{r}{|r|}\right)^n$ when $r \neq 0$.)

Exercise 13 (Some important limits). Solve the following exercises.

1. Show that $\lim_{n \rightarrow \infty} n^{1/n} = 1$.

2. Let $a > 1$. Show that $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$.

3. Let $a \in \mathbb{R}$. Show that $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$.

4. Let (x_n) be a sequence converging to 0. Define $s_n := \frac{x_1 + x_2 + \cdots + x_n}{n}$. Show that (s_n) also converges to 0.

5. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Show that $\frac{x_1 y_n + x_2 y_{n-1} + \cdots + x_n y_1}{n} \rightarrow xy$.

Example 27. Consider the sequence

$$a_n := \left(1 + \frac{1}{n}\right)^n, \quad n \in \mathbb{N}.$$

We show that (a_n) is increasing and bounded above. Consequently, it is convergent.

Step 1. (a_n) is increasing.

Consider the $(n+1)$ numbers

$$\underbrace{\frac{n+1}{n}, \frac{n+1}{n}, \dots, \frac{n+1}{n}}_{n \text{ times}}, 1.$$

By the Arithmetic Mean–Geometric Mean inequality,

$$\frac{n \left(\frac{n+1}{n} \right) + 1}{n+1} \geq \left[\left(\frac{n+1}{n} \right)^n \right]^{\frac{1}{n+1}}.$$

Since the arithmetic mean is

$$\frac{(n+1) + 1}{n+1} = 1 + \frac{1}{n+1},$$

we obtain

$$1 + \frac{1}{n+1} \geq \left(1 + \frac{1}{n} \right)^{\frac{n}{n+1}}.$$

Raising both sides to the power $n+1$ gives

$$\left(1 + \frac{1}{n+1} \right)^{n+1} \geq \left(1 + \frac{1}{n} \right)^n.$$

Hence (a_n) is increasing.

Step 2. (a_n) is bounded above.

Using the Binomial Theorem,

$$\begin{aligned} a_n &= \left(1 + \frac{1}{n} \right)^n \\ &= 1 + 1 + \frac{n(n-1)}{2!n^2} + \frac{n(n-1)(n-2)}{3!n^3} + \dots + \frac{1}{n^n} \\ &< 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(n-1)!} + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \\ &= 3. \end{aligned}$$

Thus (a_n) is bounded above. Also, we used the fact that $k! \geq 2^{k-1}$ for every $k \geq 1$, and so we have

$$\frac{1}{k!} \leq \frac{1}{2^{k-1}}.$$

Therefore, by the Monotone Convergence Theorem, (a_n) is convergent.

We denote its limit by

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n.$$

The number e is one of the most important constants in mathematics.

2.6 Sequences diverging to $\pm\infty$

Definition 17. Let (x_n) be a real sequence. We say that (x_n) diverges to $+\infty$ (or simply diverges to ∞) if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n > R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Definition 18. Let (x_n) be a real sequence. We say that (x_n) diverges to $-\infty$ if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n < R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = -\infty.$$

Example 28. The sequence (n^2) diverges to $+\infty$, whereas the sequence $(-n)$ diverges to $-\infty$. That is,

$$\lim_{n \rightarrow \infty} n^2 = +\infty \quad \text{and} \quad \lim_{n \rightarrow \infty} (-n) = -\infty.$$

Example 29. Define a sequence (x_n) by

$$x_{2k-1} = 1, \quad x_{2k} = 2k, \quad k \in \mathbb{N}.$$

The sequence (x_n) is unbounded but does not diverge to $+\infty$. Indeed, let $R = 1$. For any $N \in \mathbb{N}$, choose $n = 2N - 1$. Then $n \geq N$, but

$$x_n = x_{2N-1} = 1 \not> R.$$

Hence (x_n) does not diverge to $+\infty$.

Example 30. Let $a > 1$. Then

$$\lim_{n \rightarrow \infty} a^n = +\infty.$$

Indeed, write $a = 1 + h$ for some $h > 0$. By Bernoulli's inequality,

$$a^n = (1 + h)^n \geq 1 + nh > nh.$$

Let $R \in \mathbb{R}$ be given. By the Archimedean property, there exists $N \in \mathbb{N}$ such that

$$Nh > R.$$

Then, for every $n \geq N$,

$$a^n > nh \geq Nh > R.$$

Hence $a^n \rightarrow +\infty$.

Example 31. The sequence $((n!)^{1/n})$ diverges to $+\infty$.

Let $R > 0$ be given. Since

$$\lim_{n \rightarrow \infty} \frac{R^n}{n!} = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\frac{R^n}{n!} < 1, \quad n \geq N.$$

Thus

$$R^n < n!,$$

and therefore

$$R < (n!)^{1/n}, \quad n \geq N.$$

Hence

$$(n!)^{1/n} \rightarrow +\infty.$$

Example 32. Consider the sequence

$$x_n = (-1)^n.$$

This sequence is divergent. Moreover, it diverges neither to $+\infty$ nor to $-\infty$.

Indeed, the sequence is bounded since $-1 \leq x_n \leq 1$ for all n . Hence, it cannot diverge to $+\infty$ or to $-\infty$. On the other hand, we have already shown that (x_n) is not convergent. Therefore, (x_n) is divergent, but not divergent to $\pm\infty$.

Example 33. Define

$$x_n := \sum_{k=1}^n \frac{1}{k}, \quad n \in \mathbb{N}.$$

We show that (x_n) diverges to $+\infty$.

Let $R \in \mathbb{R}$ be given. By the Archimedean Property, choose $m \in \mathbb{N}$ such that

$$\frac{m}{2} > R.$$

Now let $N := 2^m$. For every $n \geq N$, we have

$$x_n \geq x_{2^m}.$$

Moreover,

$$\begin{aligned} x_{2^m} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots \\ &\quad + \left(\frac{1}{2^{m-1} + 1} + \cdots + \frac{1}{2^m}\right). \end{aligned}$$

Each group after the first two contains at least

$$2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2},$$

where $j = 1, 2, \dots, m$. Hence

$$x_{2^m} \geq 1 + \frac{m}{2} > R.$$

Therefore,

$$x_n \geq x_{2^m} > R, \quad n \geq N.$$

This proves that

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Exercise 14. Let

$$x_n := 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

1. Show that (x_n) is monotonically increasing.
2. Show that (x_n) is bounded above.
3. Deduce that (x_n) is convergent.
4. Show that

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

2.7 Ratio and root tests

Theorem 13 (Ratio Test). Let (u_n) be a sequence of positive real numbers such that

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell.$$

Then the following holds.

1. If $0 \leq \ell < 1$, then

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. If $\ell > 1$, then

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 3. If $\ell = 1$, no conclusion can be drawn.

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{n+1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 1.$$

Theorem 14 (Root Test). Let (u_n) be a sequence of positive real numbers such that

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell.$$

Then the following holds.

1. If $0 \leq \ell < 1$, then

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. If $\ell > 1$, then

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 4. If $\ell = 1$, no conclusion can be drawn.

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 0.$$

Example 34. Show that

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2^{n+1}/(n+1)!}{2^n/n!} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0.$$

Hence, the conclusion follows from the Ratio Test.

Example 35. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{1}{2}\right)^n} = \frac{1}{2} < 1.$$

Hence, the conclusion follows from the Root Test.

2.8 Subsequences

Definition 19. Let (r_n) be a strictly increasing sequence of natural numbers that is

$$r_{n+1} > r_n \quad \text{and} \quad r_n \geq n \quad \text{for every} \quad n \in \mathbb{N}.$$

If (x_n) is any sequence, then the sequence

$$(x_{r_n})$$

is called a subsequence of (x_n) .

One can view a subsequence of (x_n) as an infinite part of the sequence obtained by deleting some terms, while keeping the remaining terms in the same order.

Example 36. Let

$$x_n = 2^n, \quad n \in \mathbb{N}.$$

Then

$$\{2^{2n}\}_{n \in \mathbb{N}} = \{4, 16, 64, \dots\}$$

is a subsequence of (x_n) .

Here, the corresponding indices are

$$r_n = 2n, \quad n \in \mathbb{N},$$

which form a strictly increasing sequence.

Remark 5. A subsequence of (x_n) can also be written as $(x_{f(n)})$, where

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

is a strictly increasing function. Here, the n th term of the subsequence is the $f(n)$ th term of the original sequence.

Example 37. Let (x_n) be any sequence.

1. The sequence

$$(x_2, x_4, x_6, \dots)$$

is called the even subsequence of (x_n) . Here,

$$r_n = 2n, \quad n \in \mathbb{N}.$$

2. The sequence

$$(x_1, x_3, x_5, \dots)$$

is called the odd subsequence of (x_n) . Here,

$$r_n = 2n - 1, \quad n \in \mathbb{N}.$$

Theorem 15. Let (x_n) be a sequence converging to x . If (x_{r_n}) is a subsequence of (x_n) , then

$$x_{r_n} \rightarrow x.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Since (r_n) is a strictly increasing sequence of natural numbers, we have $r_n \geq n$ for every $n \in \mathbb{N}$. Hence, if $n \geq N$, then

$$r_n \geq n \geq N,$$

and therefore

$$|x_{r_n} - x| < \varepsilon.$$

Thus,

$$x_{r_n} \rightarrow x.$$

□

Exercise 15. Let (x_n) be a sequence. Suppose that both the even subsequence (x_{2n}) and the odd subsequence (x_{2n-1}) converge to the same limit x . Show that the original sequence (x_n) also converges to x .

Example 38. Show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2n}\right)^n = \sqrt{e}.$$

Solution. Let

$$x_n = \left(1 + \frac{1}{n}\right)^n.$$

We have already shown that $x_n \rightarrow e$. Hence, every subsequence of (x_n) also converges to e . In particular,

$$x_{2n} = \left(1 + \frac{1}{2n}\right)^{2n} \rightarrow e.$$

Therefore,

$$\left[\left(1 + \frac{1}{2n}\right)^n\right]^2 \rightarrow e.$$

Since $\left(1 + \frac{1}{2n}\right)^n > 0$ for every n , it follows that

$$\left(1 + \frac{1}{2n}\right)^n \rightarrow \sqrt{e}.$$

Theorem 16 (Existence of a Monotone Subsequence). *Every real sequence has a monotone subsequence.*

Theorem 17 (Bolzano–Weierstrass Theorem). *Every bounded real sequence has a convergent subsequence.*

The proofs of the above two theorems are beyond the scope of this course and are

therefore omitted. Interested students may consult a standard text on Real Analysis for their proofs.

Example 39. Assume that the sequence $(n^{1/n})$ is convergent. Show that

$$\lim_{n \rightarrow \infty} n^{1/n} = 1.$$

Solution. Let

$$n^{1/n} \longrightarrow \ell.$$

Since $((2n)^{1/(2n)})$ is a subsequence of $(n^{1/n})$, we also have

$$(2n)^{1/(2n)} \longrightarrow \ell.$$

Hence

$$((2n)^{1/(2n)})^2 \longrightarrow \ell^2.$$

On the other hand,

$$((2n)^{1/(2n)})^2 = 2^{1/n} n^{1/n}.$$

Since

$$2^{1/n} \longrightarrow 1,$$

it follows from the algebra of limits that

$$2^{1/n} n^{1/n} \longrightarrow \ell.$$

Therefore,

$$\ell^2 = \ell.$$

Since $n^{1/n} \geq 1$ for every n , we have $\ell \geq 1$. Thus,

$$\ell = 1.$$

Example 40. Show that the sequence $((-1)^n)$ is divergent.

Solution. Suppose, on the contrary, that $((-1)^n)$ converges to some real number ℓ . Then every subsequence must also converge to ℓ .

However, the even subsequence is

$$((-1)^{2n}) = (1, 1, 1, \dots),$$

which converges to 1, whereas the odd subsequence is

$$((-1)^{2n-1}) = (-1, -1, -1, \dots),$$

which converges to -1 .

Since a convergent sequence cannot have two subsequences with different limits, we obtain a contradiction. Hence

$$((-1)^n)$$

is divergent.

Exercise 16. Apply the Bolzano–Weierstrass Theorem to solve the following problems:

1. Show that the sequence $(\sin n)$ has a convergent subsequence.
2. Prove that every bounded sequence has a monotone convergent subsequence.
3. True or False: A sequence (x_n) is bounded if and only if every subsequence of (x_n) has a convergent subsequence.

3 Infinite series

In the previous chapter, we studied sequences, that is, ordered infinite collections of real numbers. A natural question now arises:

Can we meaningfully add infinitely many real numbers?

For any finite collection of real numbers

$$x_1, x_2, \dots, x_n,$$

the sum

$$x_1 + x_2 + \dots + x_n$$

is well defined. It is therefore tempting to define the “sum” of an infinite sequence

$$(x_n)$$

by writing

$$\sum_{n=1}^{\infty} x_n.$$

However, unlike finite sums, infinite sums require much greater care. If we manipulate infinitely many terms without a precise definition, we may arrive at contradictory conclusions.

For example, consider the sequence

$$x_n = (-1)^{n-1},$$

so that

$$1 - 1 + 1 - 1 + \cdots .$$

If we group the terms as

$$(1 - 1) + (1 - 1) + (1 - 1) + \cdots ,$$

then we obtain

$$0 + 0 + 0 + \cdots = 0.$$

On the other hand, if we write

$$1 + (-1 + 1) + (-1 + 1) + \cdots ,$$

then we obtain

$$1 + 0 + 0 + \cdots = 1.$$

Thus, the same infinite expression appears to have two different values:

$$1 - 1 + 1 - 1 + \cdots = 0 \quad \text{and} \quad 1 - 1 + 1 - 1 + \cdots = 1.$$

This contradiction shows that an infinite sum cannot be treated in the same way as a finite sum. We therefore need a rigorous definition of what it means for an infinite series to have a sum. This is the main objective of this chapter.

3.1 Convergence and sum of an infinite series

Definition 20. Let (a_n) be a sequence of real numbers. A formal expression of the form

$$\sum_{n=1}^{\infty} a_n,$$

or simply

$$\sum a_n \quad \text{or} \quad \sum_n a_n$$

is called an infinite series.

Definition 21 (Partial Sum). Let

$$\sum_{n=1}^{\infty} a_n$$

be an infinite series. For each $n \in \mathbb{N}$, the finite sum

$$s_n := a_1 + a_2 + \cdots + a_n$$

is called the n th partial sum of the series.

Notation. Associated with the series

$$\sum_{n=1}^{\infty} a_n,$$

we have the sequence of partial sums

$$(s_n),$$

where

$$s_n = a_1 + a_2 + \cdots + a_n, \quad n \in \mathbb{N}.$$

Thus, every infinite series naturally determines a sequence of partial sums.

Definition 22 (Convergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be convergent if the sequence of partial sums (s_n) is convergent.

Notation. If

$$s_n \longrightarrow s,$$

then we call s the *sum* of the series and write

$$\sum_{n=1}^{\infty} a_n = s.$$

Definition 23 (Divergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be divergent if the sequence of partial sums (s_n) is divergent.

Definition 24 (Absolutely Convergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be absolutely convergent if the series

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Remark 6. *If $a_n \geq 0$ for every n , then*

$$|a_n| = a_n,$$

and hence

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n.$$

Therefore, a series of non-negative terms is convergent if and only if it is absolutely convergent.

Definition 25 (Conditionally Convergent Series). *A series is said to be conditionally convergent if it is convergent but not absolutely convergent.*

Example 41. *Consider the constant sequence*

$$a_n = c, \quad n \in \mathbb{N},$$

where $c \in \mathbb{R}$ is fixed. Determine whether the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent.

Solution. *The n th partial sum is*

$$s_n = a_1 + \cdots + a_n = nc.$$

If $c = 0$, then $s_n = 0$ for every n , and hence the series converges.

Now suppose $c \neq 0$. By the Archimedean Property,

$$|nc| \longrightarrow \infty.$$

Therefore, the sequence (s_n) is divergent, and hence the series is divergent.

Thus,

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if $c = 0$.

Example 42 (Geometric Series). Let $z \in \mathbb{R}$ satisfy $|z| < 1$. Consider the infinite series

$$\sum_{n=0}^{\infty} z^n.$$

Since the n th partial sum is

$$s_n = \sum_{k=0}^n z^k = \frac{1 - z^{n+1}}{1 - z},$$

and

$$z^{n+1} \longrightarrow 0,$$

it follows that

$$s_n \longrightarrow \frac{1}{1 - z}.$$

Hence,

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}, \quad |z| < 1.$$

Example 43 (Telescoping Series). Let (a_n) and (b_n) be two sequences satisfying

$$a_n = b_{n+1} - b_n, \quad n \in \mathbb{N}.$$

Then the n th partial sum is

$$\begin{aligned} s_n &= a_1 + a_2 + \cdots + a_n \\ &= (b_2 - b_1) + (b_3 - b_2) + \cdots + (b_{n+1} - b_n) \\ &= b_{n+1} - b_1. \end{aligned}$$

Hence, the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent if and only if the sequence (b_n) is convergent. In that case,

$$\sum_{n=1}^{\infty} a_n = -b_1 + \lim_{n \rightarrow \infty} b_n.$$

Example 44. Evaluate the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Solution. Observe that

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Thus,

$$a_n = \frac{1}{n(n+1)}, \quad b_n = -\frac{1}{n}.$$

Since

$$\lim_{n \rightarrow \infty} b_n = 0,$$

the previous example gives

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = -b_1 + \lim_{n \rightarrow \infty} b_n = 1.$$

Example 45 (Harmonic Series). Recall Example 18. The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is divergent.

Example 46 (p -Series). For $p > 0$, consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Then

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. Equivalently,

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \text{convergent}, & p > 1, \\ \text{divergent}, & 0 < p \leq 1. \end{cases}$$

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 18 (Cauchy Criterion). The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for

every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n, m \geq N \implies |s_n - s_m| < \varepsilon,$$

where (s_n) denotes the sequence of partial sums.

Thus, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > m \geq N \implies |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon.$$

Proof. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. $\square$

Theorem 19. *If the series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.*

Proof. Let $s_n = \sum_{k=1}^n a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \varepsilon, \quad \forall n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore $|a_n| = |s_n - s_{n-1}| < \varepsilon$. Hence $a_n \rightarrow 0$. $\square$

Theorem 20. *Suppose that the series $\sum_{n=1}^{\infty} a_n$ converges to s . Then, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \varepsilon.$$

Proof. Let $\varepsilon > 0$ be given. Since $s_n \rightarrow s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \varepsilon$. Since $\sum_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \varepsilon.$$

This completes the proof.

Note: For some $k \in \mathbb{N}$, the sum $\sum_{n=k}^{\infty} a_n$ is called the tail of the series. $\square$

Theorem 21 (Algebra of Convergent Series). *Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two convergent series with sums A and B , respectively. Then:*

1. *The series $\sum_{n=1}^{\infty} (a_n + b_n)$ is convergent and*

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B.$$

2. *For every $\lambda \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} \lambda a_n$ is convergent and*

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 22. *If the series $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ also converges.*

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $\sigma_n = \sum_{k=1}^n |a_k|$ denote the sequences of partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} |a_n|$, respectively.

Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \varepsilon, \quad \forall n > m \geq N.$$

For $n > m \geq N$, we have

$$|s_n - s_m| = \left| \sum_{k=m+1}^n a_k \right| \leq \sum_{k=m+1}^n |a_k| = \sigma_n - \sigma_m < \varepsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for series, the series $\sum_{n=1}^{\infty} a_n$ converges. $\square$

3.2 Test of convergence

Theorem 23 (Comparison Test). *Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of non-negative real numbers. Suppose that $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then:*

1. *If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ also converges.*
2. *If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ also diverges.*

Example 47. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. Since

$$0 \leq \frac{1}{n^3} \leq \frac{1}{n^2}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is also convergent.

Example 48. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Since

$$\frac{1}{n} \leq \frac{1}{\sqrt{n}}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is also divergent.

Theorem 24 (d'Alembert's Ratio Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 49. Consider the series $\sum_{n=1}^{\infty} \frac{1}{n!}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \longrightarrow 0,$$

the Ratio Test implies that the series converges.

Example 50. Consider the series $\sum_{n=1}^{\infty} \frac{2^n}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \longrightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

Example 51. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent.

On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1} \right)^2 \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Theorem 25 (Cauchy's Root Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r.$$

Then:

1. If $0 \leq r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.
2. If $r > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
3. If $r = 1$, then the test is inconclusive.

Example 52. Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

Example 53. Consider the series $\sum_{n=1}^{\infty} 2^n$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

Example 54. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent.

On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 55. Show that the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Solution. Let $a_n = \frac{2^n n!}{n^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \left(\frac{n}{n+1} \right)^n.$$

Since $(1 + \frac{1}{n})^n \rightarrow e$, we have

$$\left(\frac{n}{n+1} \right)^n = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Example 56. Evaluate the following limits:

$$1. \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}.$$

$$2. \lim_{n \rightarrow \infty} \frac{n!}{n^n}.$$

Brief Solution.

$$1. \text{ Taking logarithms, } \log \left(\frac{(n!)^{1/n}}{n} \right) = \frac{1}{n} \sum_{k=1}^n \log \left(\frac{k}{n} \right). \text{ This is a Riemann sum for}$$

$\int_0^1 \log x \, dx = -1$. Hence

$$\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.$$

2. We have $0 \leq \frac{n!}{n^n} = \prod_{k=1}^n \frac{k}{n} \leq \left(\frac{1}{2}\right)^{\lfloor n/2 \rfloor}$. Since the right-hand side tends to 0, the Sandwich Theorem gives

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0.$$

Example 57. Determine whether the series $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ is convergent.

Solution. Let $a_n = \frac{7^{n+1}}{9^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{7^{n+2}}{9^{n+1}} \cdot \frac{9^n}{7^{n+1}} = \frac{7}{9}.$$

Since

$$\frac{7}{9} < 1,$$

the Ratio Test implies that the series $\sum_{n=1}^{\infty} \frac{7^{n+1}}{9^n}$ converges.

Theorem 26 (Abel-Pringsheim Criterion). Let (a_n) be a decreasing sequence of non-negative real numbers. If the series $\sum_{n=1}^{\infty} a_n$ converges, then $na_n \rightarrow 0$.

Proof. Since $\sum a_n$ is convergent, there exists N such that $|s_n - s_m| < \varepsilon$ for $n, m \geq N$. For $k \geq N$,

$$ka_{2k} \leq a_{k+1} + \cdots + a_{2k} = s_{2k} - s_k.$$

Thus,

$$\lim_{n \rightarrow \infty} (2n)a_{2n} = 0.$$

Since $a_{2n+1} \leq a_{2n}$, we have

$$(2n+1)a_{2n+1} \leq (2n+1)a_{2n} \leq (2n)a_{2n} + a_{2n}.$$

Now the series $\sum a_n$ is convergent, so the sequence (a_n) converges to 0, and hence the subsequence (a_{2n}) converges to zero. It follows from above that the sequence $((2n+1)a_{2n+1})$ is convergent. Using Exercise 15, we conclude that $na_n \rightarrow 0$. $\square$

Consider the series

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}.$$

Although

$$\lim_{n \rightarrow \infty} \frac{n}{n \log n} = 0,$$

the series nevertheless diverges. We will see later, using the integral test, why this series diverges.

Example 58. Show that the series $\sum_{n=1}^{\infty} \frac{1}{an+b}$, where $a, b > 0$, is divergent.

Solution. Let $a_n = \frac{1}{an+b}$. Then (a_n) is a decreasing sequence of non-negative real numbers. If the series $\sum_{n=1}^{\infty} a_n$ were convergent, then by the Abel–Pringsheim Criterion (26), we would have $na_n \rightarrow 0$.

However,

$$na_n = \frac{n}{an+b} = \frac{1}{a + \frac{b}{n}} \rightarrow \frac{1}{a} \neq 0.$$

This contradiction shows that the series $\sum_{n=1}^{\infty} \frac{1}{an+b}$ is divergent.

Exercise 17 (Abel’s Test). Assume that $\sum a_n$ is convergent and (b_n) is either increasing or decreasing, and there exist $m, M \in \mathbb{R}$ such that

$$m \leq b_n \leq M, \quad \forall n \in \mathbb{N}.$$

Then $\sum a_n b_n$ is convergent.

Definition 26. A series of the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

or

$$\sum_{n=1}^{\infty} (-1)^n a_n,$$

where $a_n \geq 0$ for all $n \in \mathbb{N}$, is called an alternating series.

Example 59. The following are alternating series:

1. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$
2. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{1}{1} + \frac{1}{4} - \frac{1}{9} + \frac{1}{16} - \cdots.$
3. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^n} = \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \cdots.$

Theorem 27 (Leibniz Test). *Let (a_n) be a decreasing sequence of non-negative real numbers such that $a_n \rightarrow 0$. Then the alternating series*

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges. Moreover, if

$$S = \sum_{n=1}^{\infty} (-1)^{n-1} a_n,$$

then

$$a_1 - a_2 \leq S \leq a_1.$$

Proof. Let

$$s_n = \sum_{k=1}^n (-1)^{k-1} a_k.$$

Observe that

$$s_{2n} = (a_1 - a_2) + \cdots + (a_{2n-1} - a_{2n}).$$

Since $a_{2n+1} \geq a_{2n+2}$,

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Hence (s_{2n}) is increasing. Moreover,

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1.$$

Thus (s_{2n}) is bounded above, and therefore converges. Let

$$s_{2n} \longrightarrow S.$$

Now,

$$s_{2n+1} = s_{2n} + a_{2n+1}.$$

Since $a_{2n+1} \rightarrow 0$, we obtain

$$s_{2n+1} \longrightarrow S.$$

Hence, both the even and odd subsequences of (s_n) converge to the same limit. Therefore,

$$s_n \longrightarrow S,$$

which proves that the series converges.

Finally,

$$a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1,$$

and letting $n \rightarrow \infty$ gives

$$a_1 - a_2 \leq S \leq a_1.$$

□

Example 60. Consider the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Since the sequence $a_n = \frac{1}{n}$ is decreasing and satisfies $a_n \rightarrow 0$, the Leibniz Test (27) implies that the series converges.

However, the series is not absolutely convergent, because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series and is divergent (see Example 18).

Thus, the alternating harmonic series is conditionally convergent.

Example 61. Show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.

Solution. Let $a_n = \frac{(-1)^{n+1}}{n}$ and $b_n = e^{-1/n}$. The series $\sum a_n$ is convergent by the Leibniz Test (27).

Moreover, (b_n) is increasing and

$$0 < b_n < 1, \quad \forall n \in \mathbb{N}.$$

Thus (b_n) is bounded and monotone. Hence, by Abel's Test (17),

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} e^{-1/n}$$

is convergent.

4 Continuity

In the previous chapters, we developed a rigorous theory of convergent sequences and studied many of their applications. It is therefore natural to ask whether convergence is preserved under a function. More precisely, if (x_n) is a sequence converging to a point a , under what conditions does the sequence $(f(x_n))$ converge to $f(a)$? Functions satisfying this property are called *continuous functions*. This sequential viewpoint provides a natural and elegant introduction to continuity and builds directly on the ideas developed earlier.

Intuitively, continuity means that small changes in the input produce only small changes in the output; in other words, there are no sudden jumps or breaks. Many quantities encountered in nature, such as temperature, speed, or pressure, vary continuously over time. Geometrically, a continuous function can often be visualized as one whose graph can be drawn from left to right without lifting the pen from the paper.

4.1 Continuous functions

Definition 27. Let $J \subseteq \mathbb{R}$, let $f : J \rightarrow \mathbb{R}$ be a function, and let $a \in J$. We say that f is continuous at a if for every sequence (x_n) in J with $x_n \rightarrow a$, we have $f(x_n) \rightarrow f(a)$. We say that f is continuous on J if it is continuous at every point of J .

Example 62. Show that every constant function is continuous.

Solution. Let $J \subseteq \mathbb{R}$ and let $f : J \rightarrow \mathbb{R}$ be the constant function defined by $f(x) = c$ for all $x \in J$.

Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \rightarrow a$. Since $f(x_n) = c$ for every n , the sequence $(f(x_n))$ is constant. Hence,

$$f(x_n) \rightarrow c = f(a).$$

Therefore, f is continuous at a . As a was arbitrary, we conclude that f is continuous on J .

Example 63. Show that the identity function $f : J \rightarrow \mathbb{R}$, defined by $f(x) = x$ for all $x \in J$, is continuous.

Solution. Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \rightarrow a$. Since $f(x_n) = x_n$ for every n , we have

$$f(x_n) = x_n \rightarrow a = f(a).$$

Hence, f is continuous at a . Since a was arbitrary, f is continuous on J .

More generally, for every $n \in \mathbb{N}$, the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^n$ is continuous.

Example 64. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then f is not continuous at 0.

Solution. Consider the sequence $x_n = \frac{1}{n}$. Then $x_n \in \mathbb{Q}$ and $x_n \rightarrow 0$. Hence

$$f(x_n) = 1 \longrightarrow 1 = f(0).$$

Now let

$$y_n = \frac{\sqrt{2}}{n}.$$

Since $\sqrt{2}$ is irrational, y_n is irrational for every n . Also, $y_n \rightarrow 0$. Therefore,

$$f(y_n) = 0 \longrightarrow 0 \neq 1 = f(0).$$

Thus, by the sequential criterion for continuity, f is not continuous at 0.

Using the density of the rational and irrational numbers in $\mathbb{R}$, together with the Sandwich Theorem, one can similarly show that f is not continuous at any point of $\mathbb{R}$.

Example 65. Consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

We claim that f is continuous at every $a \neq 0$ but is not continuous at 0.

Indeed, let $a \neq 0$ and let (x_n) be a sequence converging to a . Since a is either positive or negative, there exists $N \in \mathbb{N}$ such that x_n has the same sign as a for all $n \geq N$. Consequently, $f(x_n) = f(a)$ for all sufficiently large n , and hence

$$f(x_n) \longrightarrow f(a).$$

Thus f is continuous at every $a \neq 0$.

Now let $a = 0$. Consider the sequences

$$x_n = \frac{1}{n}, \quad y_n = -\frac{1}{n}.$$

Both converge to 0, but

$$f(x_n) = 1 \longrightarrow 1, \quad f(y_n) = 0 \longrightarrow 0.$$

Since the limits are different, f is not continuous at 0.

Example 66. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \alpha, & x < 0, \\ ax^2 - bx + c, & x \geq 0. \end{cases}$$

Determine the value of α for which f is continuous at 0.

Solution. Suppose that f is continuous at 0. Consider the sequence

$$x_n = -\frac{1}{n}.$$

Then $x_n \rightarrow 0$ and

$$f(x_n) = \alpha \longrightarrow f(0) = c.$$

Hence,

$$\alpha = c.$$

Conversely, assume that $\alpha = c$. Let (x_n) be any sequence with $x_n \rightarrow 0$. If $x_n < 0$, then

$$f(x_n) = c = f(0).$$

If $x_n \geq 0$, then

$$f(x_n) - f(0) = ax_n^2 - bx_n \longrightarrow 0,$$

by the algebra of convergent sequences. Therefore,

$$f(x_n) \longrightarrow f(0).$$

Hence, f is continuous at 0. Thus, f is continuous at 0 if and only if

$$\boxed{\alpha = c.}$$

Theorem 28 (Algebra of Continuous Functions). Let $J \subseteq \mathbb{R}$, let $f, g : J \rightarrow \mathbb{R}$ be continuous at $a \in J$, and let $\alpha \in \mathbb{R}$. Then:

1. $f + g$ is continuous at a .
2. αf is continuous at a .
3. fg is continuous at a .
4. If $g(a) \neq 0$, then there exists $\delta > 0$ such that $g(x) \neq 0$ for all $x \in (a - \delta, a + \delta) \cap J$.
Then $\frac{1}{g}$ is continuous at a .
5. $|f|$ is continuous at a .
6. The functions $\max\{f, g\}$ and $\min\{f, g\}$ are continuous at a .

7. If $K \subseteq \mathbb{R}$, $h : K \rightarrow \mathbb{R}$ is continuous at $f(a)$, and $f(J) \subseteq K$, then the composition $h \circ f$ is continuous at a .

4.2 $\varepsilon - \delta$ definition of continuity

Before giving the formal definition of continuity, let us first understand its basic idea. Suppose we want the value of a function to be close to a desired output $f(a)$. A natural question is:

How close should the input x be to a so that the output $f(x)$ is as close as we want to $f(a)$?

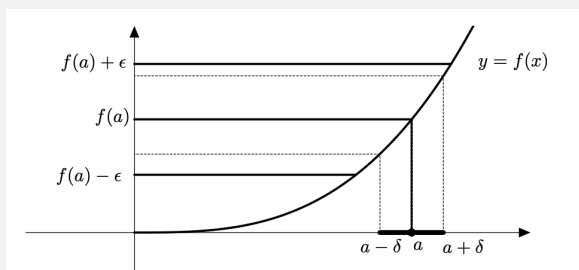
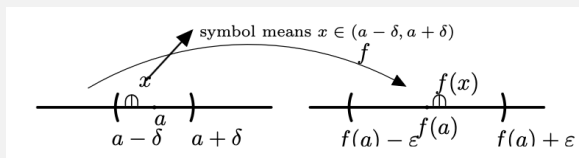
The answer depends on how much error we are willing to tolerate in the output. If we specify a tolerance $\varepsilon > 0$ for the output, continuity guarantees that there exists a tolerance $\delta > 0$ for the input such that whenever x is within δ of a , the corresponding value $f(x)$ is within ε of $f(a)$.

This simple idea forms the basis of the classical ε - δ definition of continuity.

Definition 28. Let $J \subseteq \mathbb{R}$, let $f : J \rightarrow \mathbb{R}$ be a function, and let $a \in J$. We say that f is continuous at a if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

We say that f is continuous on J if it is continuous at every point of J .



Theorem 29. *The sequential definition of continuity (Definition 27) and the ε - δ definition (Definition 28) are equivalent.*

Proof. We prove both implications.

(ε - δ continuity $\Rightarrow$ sequential continuity). Suppose f is continuous at a in the sense of Definition 28. Let (x_n) be a sequence in J such that $x_n \rightarrow a$. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad |x - a| < \delta \quad \Longrightarrow \quad |f(x) - f(a)| < \varepsilon.$$

Since $x_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |x_n - a| < \delta.$$

Hence

$$n \geq N \quad \Longrightarrow \quad |f(x_n) - f(a)| < \varepsilon.$$

Thus $f(x_n) \rightarrow f(a)$, so f is continuous at a in the sense of Definition 27.

(Sequential continuity $\Rightarrow$ ε - δ continuity). Suppose f is continuous at a in the sense of Definition 27. Assume, to the contrary, that the ε - δ condition fails. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, there is $x \in J$ satisfying

$$|x - a| < \delta, \quad |f(x) - f(a)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose $x_n \in J$ such that

$$|x_n - a| < \frac{1}{n}, \quad |f(x_n) - f(a)| \geq \varepsilon_0.$$

Then $x_n \rightarrow a$. By sequential continuity, $f(x_n) \rightarrow f(a)$. Hence there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |f(x_n) - f(a)| < \varepsilon_0,$$

contradicting $|f(x_n) - f(a)| \geq \varepsilon_0$ for every n .

Therefore, the ε - δ condition holds, and the two definitions are equivalent. $\square$

Example 67. *Show that the identity function*

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x,$$

is continuous on $\mathbb{R}$.

Solution. Let $a \in \mathbb{R}$ and let $\varepsilon > 0$ be given. We have

$$|f(x) - f(a)| = |x - a|.$$

Thus, it is natural to choose

$$\delta = \varepsilon.$$

Then, whenever $|x - a| < \delta$, we obtain

$$|f(x) - f(a)| = |x - a| < \delta = \varepsilon.$$

Hence, f is continuous at a . Since a was arbitrary, f is continuous on $\mathbb{R}$.

Example 68. Suppose $R > 0$. Show that the function

$$f : [-R, R] \rightarrow \mathbb{R}, \quad f(x) = x^2,$$

is continuous.

Solution. Let $a \in [-R, R]$ and let $\varepsilon > 0$ be given. Observe that

$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a| \leq (|x| + |a|)|x - a| \leq 2R|x - a|.$$

Choose

$$\delta = \frac{\varepsilon}{2R}.$$

Then, whenever $|x - a| < \delta$, we have

$$|f(x) - f(a)| \leq 2R|x - a| < 2R\delta = \varepsilon.$$

Hence, f is continuous on $[-R, R]$.

Exercise 18. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = x^2$, is continuous on $\mathbb{R}$ using the ε - δ definition. Hint. Observe that $|x^2 - a^2| = |x - a||x + a|$. To estimate $|x + a|$, first assume $\delta \leq 1$. Then, whenever $|x - a| < \delta$, we have

$$|x| \leq |x - a| + |a| < 1 + |a|,$$

and hence

$$|x + a| \leq |x| + |a| < 1 + 2|a|.$$

Now, choose $\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|a|} \right\}$.

Theorem 30. Let $J \subseteq \mathbb{R}$ and let $f : J \rightarrow \mathbb{R}$ be continuous at $c \in J$. If $f(c) \neq 0$, then there exists $\delta > 0$ such that $|f(x)| > \frac{|f(c)|}{2}$ for all $x \in (c - \delta, c + \delta) \cap J$.

In particular, if $f(c) > 0$, then there exists $\delta > 0$ such that $f(x) > \frac{f(c)}{2}$ for all $x \in (c - \delta, c + \delta) \cap J$.

Proof. Choose $\varepsilon = \frac{|f(c)|}{2}$. Since f is continuous at c , there exists $\delta > 0$ such that

$|f(x) - f(c)| < \varepsilon$ whenever $x \in (c - \delta, c + \delta) \cap J$.

Hence,

$$|f(x)| \geq |f(c)| - |f(x) - f(c)| > |f(c)| - \frac{|f(c)|}{2} = \frac{|f(c)|}{2}.$$

If $f(c) > 0$, then $|f(x) - f(c)| < \frac{f(c)}{2}$ implies $f(x) > \frac{f(c)}{2}$. □

4.3 Properties of continuous functions

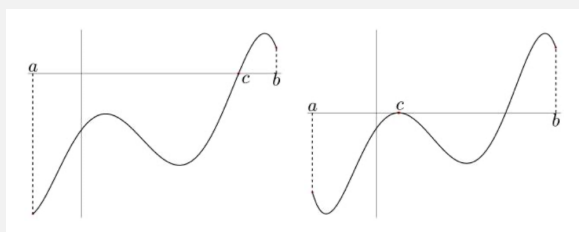
Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function such that $f(a)$ and $f(b)$ have opposite signs. Draw the graphs of a few such functions, considering both the cases $f(a) > 0 > f(b)$ and $f(a) < 0 < f(b)$.

What do you observe? Does the graph always intersect the x -axis? If so, can you explain why this must happen? In other words, must there exist a point $c \in (a, b)$ such that $f(c) = 0$?

The answer is affirmative, and it is one of the most important consequences of continuity. This result is known as the *Intermediate Value Theorem*.

Theorem 31 (Intermediate Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function such that $f(a) < 0 < f(b)$. Then there exists a point $c \in (a, b)$ satisfying $f(c) = 0$.*

Sketch of the proof. The idea is illustrated in the following figure.



We look for the *first* point where the graph meets the x -axis. To do this, consider the set

$$E = \{x \in [a, b] : f(y) \leq 0 \text{ for every } y \in [a, x]\}.$$

Clearly, E is nonempty (since $a \in E$) and bounded above by b . By the Completeness Property of $\mathbb{R}$, the set E has a least upper bound; let

$$c = \sup E.$$

Using the continuity of f , one shows that neither $f(c) < 0$ nor $f(c) > 0$ is possible. Therefore,

$$f(c) = 0.$$

Hence, there exists $c \in (a, b)$ such that $f(c) = 0$. □

Theorem 32 (Intermediate Value Theorem (Standard Version)). *Let $g : [a, b] \rightarrow \mathbb{R}$ be a continuous function, and let λ be a real number between $g(a)$ and $g(b)$ that is $\min\{g(a), g(b)\} < \lambda < \max\{g(a), g(b)\}$. Then there exists a point $c \in (a, b)$ such that*

$$g(c) = \lambda.$$

Proof. Without loss of generality, assume that $g(a) < \lambda < g(b)$. Define

$$f(x) = g(x) - \lambda.$$

Since g is continuous on $[a, b]$, so is f . Moreover,

$$f(a) = g(a) - \lambda < 0 \quad \text{and} \quad f(b) = g(b) - \lambda > 0.$$

Hence, by the Intermediate Value Theorem 31, there exists $c \in (a, b)$ such that

$$f(c) = 0.$$

Therefore,

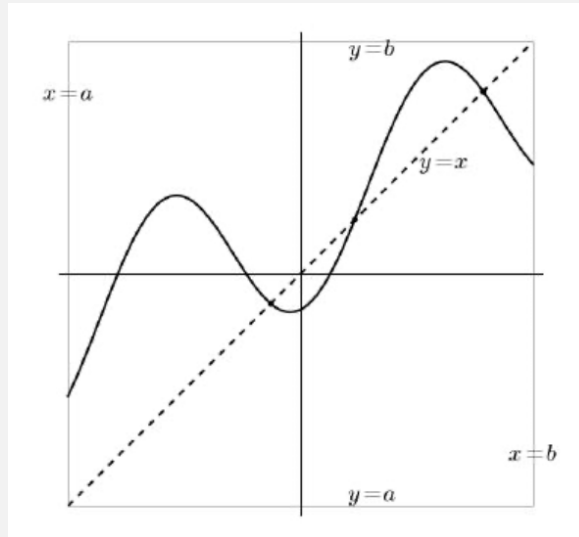
$$g(c) = \lambda.$$

□

Exercise 19. *Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function whose range is contained in $\mathbb{Z}$, then f is constant.*

Theorem 33 (Fixed Point Theorem). *Let $f : [a, b] \rightarrow [a, b]$ be a continuous function. Then there exists $c \in [a, b]$ such that $f(c) = c$. Such a point c is called a fixed point of f .*

Proof. Define $g(x) = f(x) - x$. Since g is continuous on $[a, b]$, we have $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$. If either $g(a) = 0$ or $g(b) = 0$, then we are done. Otherwise, $g(a) > 0 > g(b)$. Hence, by the Intermediate Value Theorem 31, there exists $c \in (a, b)$ such that $g(c) = 0$. Therefore, $f(c) = c$.



□

Example 69. Show that the equation $xe^x = 1$ has a solution in the interval $(0, 1)$.

Solution. Define $f : [0, 1] \rightarrow \mathbb{R}$ by $f(x) = xe^x - 1$. Since f is continuous on $[0, 1]$, we have $f(0) = -1 < 0$ and $f(1) = e - 1 > 0$. Hence, by the Intermediate Value Theorem, there exists $c \in (0, 1)$ such that $f(c) = 0$. Therefore, $ce^c = 1$.

Example 70. Let $f : [0, 2\pi] \rightarrow \mathbb{R}$ be a continuous function satisfying $f(0) = f(2\pi)$. Show that there exists $x \in [0, \pi]$ such that $f(x) = f(x + \pi)$.

Solution. Define $g : [0, \pi] \rightarrow \mathbb{R}$ by $g(x) = f(x) - f(x + \pi)$. Since f is continuous, so is g . Moreover,

$$g(0) = f(0) - f(\pi) \quad \text{and} \quad g(\pi) = f(\pi) - f(2\pi) = f(\pi) - f(0) = -g(0).$$

If $g(0) = 0$, then $f(0) = f(\pi)$ and we are done. Otherwise, $g(0)$ and $g(\pi)$ have opposite signs. Hence, by the Intermediate Value Theorem, there exists $c \in (0, \pi)$ such that $g(c) = 0$. Therefore, $f(c) = f(c + \pi)$.

Example 71. Show that every real polynomial of odd degree defines an onto function from $\mathbb{R}$ to $\mathbb{R}$.

Solution. Let $p : \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial of odd degree, and let $y \in \mathbb{R}$ be arbitrary. We wish to show that there exists $x \in \mathbb{R}$ such that $p(x) = y$.

Define the function $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = p(x) - y$. Since f is continuous, it suffices to show that f changes sign.

As p has odd degree, we have $\lim_{x \rightarrow \infty} p(x) = \infty$ and $\lim_{x \rightarrow -\infty} p(x) = -\infty$, or vice versa, depending on the sign of the leading coefficient. Hence, there exist $a, b \in \mathbb{R}$ such that

$f(a) < 0 < f(b)$.

By the Intermediate Value Theorem, there exists $c \in (a, b)$ such that $f(c) = 0$. Therefore, $p(c) = y$.

Since y was arbitrary, $p : \mathbb{R} \rightarrow \mathbb{R}$ is onto.

Definition 29. Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. We say that f is bounded on A if there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Example 72. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x$ is not bounded. Indeed, for every $M > 0$, choosing $x = M + 1$ gives $f(x) = M + 1 > M$. Hence, f is unbounded.

Example 73. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin x$ is bounded. In fact, $|f(x)| \leq 1$ for every $x \in \mathbb{R}$.

Theorem 34. [Weierstrass Theorem] Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then f is bounded.

Sketch of the proof. The proof is based on the same least upper bound argument used in the proof of the Intermediate Value Theorem.

Consider the set of all points $x \in [a, b]$ such that f is bounded on the interval $[a, x]$. Since f is continuous, it is bounded in a neighborhood of a . Thus, this set is nonempty and bounded above by b . Let c be its least upper bound.

If $c < b$, then continuity of f at c implies that f is bounded in a neighborhood of c . Consequently, f is bounded on a larger interval $[a, c_1]$ for some $c_1 > c$, contradicting the definition of c . Hence, $c = b$, and therefore f is bounded on $[a, b]$. $\square$

Theorem 35. [Extreme Value Theorem] Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then there exist points $x_{\min}, x_{\max} \in [a, b]$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ for every $x \in [a, b]$.

Proof. By the Weierstrass Theorem 34, the set $A = \{f(x) : x \in [a, b]\}$ is bounded. Let $M = \sup A$.

For each $n \in \mathbb{N}$, choose $x_n \in [a, b]$ such that $M - \frac{1}{n} < f(x_n) \leq M$. Then $f(x_n) \rightarrow M$. Since (x_n) is a bounded sequence, the Bolzano–Weierstrass Theorem yields a subsequence (x_{n_k}) converging to some $x \in [a, b]$. By continuity, $f(x_{n_k}) \rightarrow f(x)$. Hence, $f(x) = M$.

Similarly, if $m = \inf A$, there exists $y \in [a, b]$ such that $f(y) = m$. Therefore, f attains both its maximum and minimum values on $[a, b]$. $\square$

Example 74. Show that there is no continuous function $f : [0, 1] \rightarrow (0, \infty)$ which is onto.

Solution. Suppose such a function exists. Since f is continuous on the closed interval $[0, 1]$, the Weierstrass Theorem implies that f is bounded. Hence there exists $M > 0$ such that $f(x) \leq M$ for every $x \in [0, 1]$. Therefore, no number greater than M belongs to the range of f , which contradicts the assumption that f is onto.

Example 75. Show that there is no continuous function $f : [a, b] \rightarrow (0, 1)$ which is onto.

Solution. Suppose such a function exists. By the Extreme Value Theorem, there exist $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}), \quad \forall x \in [a, b].$$

Since the range of f is $(0, 1)$, we have $0 < f(x_{\min})$ and $f(x_{\max}) < 1$. Hence, neither 0 nor 1 belongs to the range of f , contradicting the assumption that f is onto.

4.4 Integral test for series

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function satisfying $\alpha \leq f(x) \leq \beta$ for all $x \in [a, b]$. Then

$$\alpha(b - a) \leq \int_a^b f(x) dx \leq \beta(b - a).$$

The above estimate has a simple geometric interpretation when f is non-negative: the area under the graph of f lies between the areas of the rectangles of heights α and β over the interval $[a, b]$.

Definition 30. Let $J \subseteq \mathbb{R}$ be an interval and $f : J \rightarrow \mathbb{R}$ be a function. We say that f is increasing on J if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2)$$

for all $x_1, x_2 \in J$.

Similarly, f is said to be decreasing on J if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2)$$

for all $x_1, x_2 \in J$.

Theorem 36 (Integral Test). *Let $f : [1, \infty) \rightarrow [0, \infty)$ be a continuous and decreasing function. Suppose that $a_n = f(n)$ for all $n \in \mathbb{N}$. Then:*

1. *If $\lim_{R \rightarrow +\infty} \int_1^R f(x) dx \rightarrow c$ where $-\infty < c < +\infty$ i.e. integral converges, then the series $\sum_{n=1}^{\infty} a_n$ converges.*
2. *If $\lim_{R \rightarrow +\infty} \int_1^R f(x) dx \rightarrow +\infty$ i.e. the integral diverges, then the series $\sum_{n=1}^{\infty} a_n$ diverges.*

Proof. For each $n \geq 2$, since f is decreasing, we have

$$a_n \leq \int_{n-1}^n f(x) dx \leq a_{n-1}.$$

Summing these inequalities from $k = 2$ to n , we obtain

$$\sum_{k=2}^n a_k \leq \int_1^n f(x) dx \leq \sum_{k=1}^{n-1} a_k.$$

Suppose first that $\int_1^{\infty} f(x) dx$ converges that is $\lim_{R \rightarrow +\infty} \int_1^R f(x) dx \rightarrow c$ where $-\infty < c < +\infty$. Then the sequence $b_n = \int_1^n f(x) dx$ is bounded. Since

$$\sum_{k=2}^n a_k \leq b_n,$$

the sequence of partial sums $(\sum_{k=2}^n a_k)$ is bounded above. As it is also increasing, it converges. Hence $\sum_{n=1}^{\infty} a_n$ converges.

Conversely, suppose that $\int_1^{\infty} f(x) dx$ diverges that is $\lim_{R \rightarrow +\infty} \int_1^R f(x) dx \rightarrow +\infty$. Then $b_n \rightarrow \infty$. Since

$$b_n \leq \sum_{k=1}^{n-1} a_k,$$

the sequence of partial sums of $\sum_{n=1}^{\infty} a_n$ is unbounded. Therefore, the series diverges. $\square$

Example 76 (p -Series). *Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, where $p > 0$. Applying the Integral Test to the function $f(x) = x^{-p}$, we obtain*

$$\int_1^{\infty} \frac{dx}{x^p} := \lim_{R \rightarrow +\infty} \int_1^R \frac{dx}{x^p} = \begin{cases} \frac{1}{p-1}, & p > 1, \\ \infty, & 0 < p \leq 1. \end{cases}$$

Hence,

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if $p > 1$ and diverges if $0 < p \leq 1$.

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Example 77. Show that the series $\sum_{n=2}^{\infty} \frac{\log n}{n^p}$ converges for $p > 1$.

Solution. Consider the function $f(x) = \frac{\log x}{x^p}$, defined for $x \geq 2$. For $p > 1$, the function f is positive, continuous, and decreasing for all sufficiently large x . By the Integral Test, it is enough to examine

$$\int_2^{\infty} \frac{\log x}{x^p} dx = \lim_{R \rightarrow +\infty} \int_2^R \frac{\log x}{x^p} dx.$$

Using integration by parts with $u = \log x$ and $dv = x^{-p} dx$, we obtain

$$\int \frac{\log x}{x^p} dx = \frac{x^{1-p} \log x}{1-p} - \frac{x^{1-p}}{(1-p)^2} + C.$$

Since $p > 1$, we have $x^{1-p} \rightarrow 0$ and $x^{1-p} \log x \rightarrow 0$ as $x \rightarrow \infty$. Hence $\int_2^{\infty} \frac{\log x}{x^p} dx$ converges. Therefore, by the Integral Test, $\sum_{n=2}^{\infty} \frac{\log n}{n^p}$ converges.

4.5 Limits

In the previous section, we studied continuity using the value of the function at a point. However, many important questions depend not on the value of the function at the point itself, but on the behavior of the function *near* that point.

For example, consider the function $f(x) = \frac{x^2 - 1}{x - 1}$ for $x \neq 1$. Although $f(1)$ is not defined, the values of $f(x)$ for x close to 1 appear to approach the number 2. This suggests that a function may exhibit a predictable behavior near a point even when it is not defined there.

Similarly, when studying discontinuous functions, we often wish to compare the behavior of the function from the left and the right of a point. These ideas naturally lead to the notion of the *limit* of a function, which describes the value that $f(x)$ approaches as x approaches a given point.

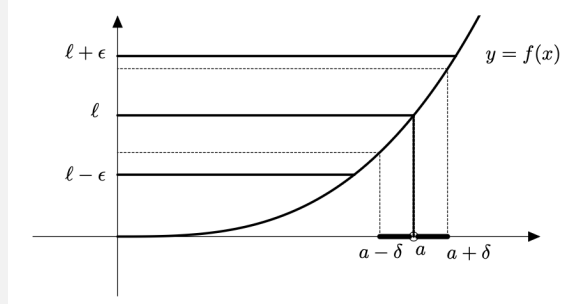
Definition 31. Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a} f(x) = \ell$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \implies |f(x) - \ell| < \varepsilon.$$

In this case, we say that ℓ is the limit of f at a .



Example 78. Show that $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$.

Solution. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. If $0 < |x - 1| < \delta$, then

$$\left| \frac{x^2 - 1}{x - 1} - 2 \right| = |x + 1 - 2| = |x - 1| < \delta = \varepsilon.$$

Hence, $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$.

Example 79. Show that $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

Solution. Let $\varepsilon > 0$ be given and choose $\delta = \varepsilon$. If $0 < |x| < \delta$, then

$$\left| x \sin\left(\frac{1}{x}\right) \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| < \delta = \varepsilon.$$

Hence, $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

Theorem 37. The limit of a function, if it exists, is unique.

Proof. Suppose $\lim_{x \rightarrow a} f(x) = \ell_1$ and $\lim_{x \rightarrow a} f(x) = \ell_2$.

Assume, to the contrary, that $\ell_1 \neq \ell_2$. Choose $\varepsilon = \frac{|\ell_1 - \ell_2|}{3} > 0$. Then there exist

$\delta_1, \delta_2 > 0$ such that

$$0 < |x - a| < \delta_j \implies |f(x) - \ell_j| < \varepsilon, \quad j = 1, 2.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x - a| < \delta$, then

$$|f(x) - \ell_1| < \varepsilon \quad \text{and} \quad |f(x) - \ell_2| < \varepsilon.$$

Hence,

$$|\ell_1 - \ell_2| \leq |\ell_1 - f(x)| + |f(x) - \ell_2| < 2\varepsilon = \frac{2}{3}|\ell_1 - \ell_2|,$$

which is impossible. Therefore, $\ell_1 = \ell_2$. □

Theorem 38 (Algebra of Finite Limits). *Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f, g : J \setminus \{a\} \rightarrow \mathbb{R}$ be functions. Assume that*

$$\lim_{x \rightarrow a} f(x) = \ell_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \ell_2,$$

where $\ell_1, \ell_2 \in \mathbb{R}$. Let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = \ell_1 + \ell_2$.
2. $\lim_{x \rightarrow a} (\alpha f(x)) = \alpha \ell_1$.
3. $\lim_{x \rightarrow a} (f(x)g(x)) = \ell_1 \ell_2$.
4. If $\ell_1 \neq 0$, then there exists $\delta > 0$ such that $f(x) \neq 0$ whenever $0 < |x - a| < \delta$.
Moreover,

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{\ell_1}.$$

Theorem 39 (Composition Rule for Limits). *Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ satisfy*

$$\lim_{x \rightarrow a} f(x) = \alpha.$$

Suppose that g is defined on an interval containing α and is continuous at α . Then

$$\lim_{x \rightarrow a} (g \circ f)(x) = g(\alpha).$$

Definition 32 (Right-Hand Limit). *Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$ be not the*

right endpoint of J , and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a^+} f(x) = \ell$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < x - a < \delta \implies |f(x) - \ell| < \varepsilon.$$

In this case, ℓ is called the right-hand limit of f at a .

Definition 33 (Left-Hand Limit). Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$ be not the left endpoint of J , and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a^-} f(x) = \ell$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < a - x < \delta \implies |f(x) - \ell| < \varepsilon.$$

In this case, ℓ is called the left-hand limit of f at a .

Example 80. Let

$$f(x) = \begin{cases} x^2, & x \geq 0; \\ 1, & x < 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^+} f(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = 1.$$

Theorem 40. Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$ be neither the left endpoint nor the right endpoint of J , and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. Then

$$\lim_{x \rightarrow a} f(x) = \ell$$

if and only if

$$\lim_{x \rightarrow a^-} f(x) = \ell \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = \ell.$$

Theorem 41. Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$ be neither the left endpoint nor the right endpoint of J , and let $f : J \rightarrow \mathbb{R}$ be a function. Then f is continuous at a if and only if

$$\lim_{x \rightarrow a^-} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = f(a).$$

Definition 34. Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$ be neither the left endpoint nor the right endpoint of J , and let $f : J \rightarrow \mathbb{R}$ be a function. We say that f is discontinuous at a if at least one of the following holds:

1. $\lim_{x \rightarrow a^-} f(x)$ does not exist;
2. $\lim_{x \rightarrow a^+} f(x)$ does not exist;
3. $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$; *Jump discontinuity*
4. $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \neq f(a)$. *Removable discontinuity*

Example 81. Consider the function

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & x < 0, \\ 0, & x \geq 0. \end{cases}$$

Then $\lim_{x \rightarrow 0^-} f(x)$ does not exist.

Example 82. Consider the function

$$f(x) = \begin{cases} 0, & x \leq 0, \\ \sin\left(\frac{1}{x}\right), & x > 0. \end{cases}$$

Then $\lim_{x \rightarrow 0^+} f(x) = 0$, does not exist.

Example 83. Consider the function

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^-} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = 1.$$

Since the left-hand and right-hand limits are different, f is discontinuous at 0.

Example 84. Consider the function

$$f(x) = \begin{cases} 1, & x \neq 0, \\ 2026, & x = 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1,$$

but $f(0) = 2026$. Hence f is discontinuous at 0.

Example 85. Consider the function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Since every interval contains both rational and irrational numbers, neither

$$\lim_{x \rightarrow 0^-} f(x) \quad \text{nor} \quad \lim_{x \rightarrow 0^+} f(x)$$

exists. Hence f is discontinuous at 0. Indeed, the function is discontinuous at every point of $\mathbb{R}$.

Example 86. Define

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then f is continuous only at $x = 0$ and is discontinuous at every other point of $\mathbb{R}$.

Theorem 42 (Sequential Characterization of Limits). Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. Then

$$\lim_{x \rightarrow a} f(x) = \ell$$

if and only if for every sequence (x_n) in $J \setminus \{a\}$ satisfying

$$x_n \rightarrow a,$$

we have

$$f(x_n) \rightarrow \ell.$$

Example 87. Show that

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

Solution. Let (x_n) be any sequence such that $x_n \rightarrow 2$. Then, by the algebra of convergent sequences,

$$x_n^2 + 1 \rightarrow 2^2 + 1 = 5.$$

Hence, by the Sequential Characterization of Limits,

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

Exercise 20. Let

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Show that

$$\lim_{x \rightarrow a} f(x)$$

exists only for $a = 0$.

Hint. Use the density of the rational and irrational numbers in $\mathbb{R}$. Alternatively, use the Sequential Characterization of Limits by considering one sequence of rational numbers and one sequence of irrational numbers, both converging to a .

Exercise 21. Find the value of c for which the function

$$f(x) = \begin{cases} cx + 1, & x < 1, \\ x^2, & x \geq 1 \end{cases}$$

is continuous at $x = 1$.

Definition 35 (Infinite Limits). Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function.

1. We say that

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \implies f(x) > M.$$

2. We say that

$$\lim_{x \rightarrow a} f(x) = -\infty$$

if for every $M < 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \implies f(x) < M.$$

Example 88. Show that $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty$.

Solution. Let $M > 0$ be given. Choose $\delta = \frac{1}{\sqrt{M}}$. If $0 < |x-2| < \delta$, then

$$\frac{1}{(x-2)^2} > \frac{1}{\delta^2} = M.$$

Hence, by the definition of an infinite limit, $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty$.

Definition 36 (Limits at Infinity). Let $A \in \mathbb{R}$ and let $f : (A, +\infty) \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow +\infty} f(x) = \ell$$

if for every $\varepsilon > 0$, there exists $M > A$ such that

$$x > M \implies |f(x) - \ell| < \varepsilon.$$

Similarly, if $f : (-\infty, A) \rightarrow \mathbb{R}$, we say that

$$\lim_{x \rightarrow -\infty} f(x) = \ell$$

if for every $\varepsilon > 0$, there exists $M < A$ such that

$$x < M \implies |f(x) - \ell| < \varepsilon.$$

Example 89. Show that $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$.

Solution. Let $\varepsilon > 0$ be given. Choose $M = \frac{1}{\varepsilon}$. If $x > M$, then

$$\left| \frac{1}{x} - 0 \right| = \frac{1}{x} < \frac{1}{M} = \varepsilon.$$

Hence, by the definition of limits at infinity, $\lim_{x \rightarrow +\infty} \frac{1}{x} = 0$.

Definition 37 (Infinite Limits at Infinity). Let $A \in \mathbb{R}$ and let $f : (A, +\infty) \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow +\infty} f(x) = +\infty$$

if for every $M > 0$, there exists $K > A$ such that

$$x > K \implies f(x) > M.$$

Similarly, let $f : (-\infty, A) \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow -\infty} f(x) = -\infty$$

if for every $N < 0$, there exists $K < A$ such that

$$x < K \implies f(x) < N.$$

Example 90. Show that $\lim_{x \rightarrow +\infty} x^2 = +\infty$.

Solution. Let $M > 0$ be given. Choose $K = \sqrt{M}$. If $x > K$, then $x^2 > K^2 = M$. Hence, by the definition of an infinite limit at infinity, $\lim_{x \rightarrow +\infty} x^2 = +\infty$.

5 Differentiation

Continuity tells us that a function has no sudden “jumps,” but it does not describe the *rate at which* the function changes. This is exactly the role of derivatives: they answer the question,

How fast does $f(x)$ change as x changes?

Think of driving a car. As time passes, the distance you have traveled increases, so we can view distance as a function of time. Now, what does the speedometer show? It tells you your speed at that exact moment—in other words, how quickly your distance is changing with respect to time.

That number the speedometer shows is exactly what a derivative represents: the rate of change at a given instant. A derivative tells us, “if I make a tiny change in the input, how much does the output change?” Mathematically, we express this idea with a limit, and that limit gives the precise definition of the derivative.

5.1 Differentiability of functions

Definition 38. Let (a, b) be an interval and $f : (a, b) \rightarrow \mathbb{R}$ be a function. Let $c \in (a, b)$. We say f is differentiable at c if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists. This limit is called the derivative of f at c and is denoted as $f'(c)$.

Corresponding to a function f , we define its derivative f' as the function whose domain consists of all points x at which the derivative of f exists.

Example 91. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x + e$. Then f is differentiable at every point of $\mathbb{R}$. For any $c \in \mathbb{R}$, we compute

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} \frac{3x + e - (3c + e)}{x - c} = \lim_{x \rightarrow c} \frac{3(x - c)}{x - c} = 3.$$

Thus, the derivative exists and

$$f'(c) = 3 \quad \text{for all } c \in \mathbb{R}.$$

Example 92. Let

$$f(x) = \begin{cases} x + 1, & x \geq 0, \\ x, & x < 0. \end{cases}$$

Here f is not differentiable at $x = 0$. First note $f(0) = 0 + 1 = 1$. Consider the difference quotient

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x) - 1}{x}, \quad x \neq 0.$$

Now for $x > 0$ then $f(x) = x + 1$, hence

$$\lim_{x \rightarrow 0^+} \frac{f(x) - 1}{x} = \lim_{x \rightarrow 0^+} \frac{x + 1 - 1}{x} = 1.$$

If $x < 0$ then $f(x) = x$, hence

$$\frac{f(x) - 1}{x} = \frac{x - 1}{x} = 1 - \frac{1}{x}.$$

As $x \rightarrow 0^-$ we have $1/x \rightarrow -\infty$, so

$$\lim_{x \rightarrow 0^-} \frac{f(x) - 1}{x} = +\infty,$$

hence, the left-hand limit diverges.

Since the left and right limits of the difference quotient at 0 are different (indeed, the left-hand limit diverges), the derivative

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$$

does not exist. Hence f is not differentiable at $x = 0$.

Theorem 43. If f is differentiable at $c \in (a, b)$, then f is continuous at c .

Proof. Suppose

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = L$$

exists and L is finite as f is differentiable at c . Then, for $x \neq c$,

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} (x - c).$$

Taking the limit as $x \rightarrow c$, the first factor tends to L , while the second tends to 0. Hence

$$\lim_{x \rightarrow c} (f(x) - f(c)) = L \cdot 0 = 0.$$

Therefore,

$$\lim_{x \rightarrow c} f(x) = f(c),$$

which shows f is continuous at c . □

Example 93. The converse does not hold: continuity need not imply differentiability. Let $f(x) = |x|$. Then f is continuous everywhere. At $c = 0$,

$$\lim_{x \rightarrow 0^+} \frac{|x| - |0|}{x - 0} = 1, \quad \lim_{x \rightarrow 0^-} \frac{|x| - |0|}{x - 0} = -1.$$

The one-sided limits differ, so $f'(0)$ does not exist.

Example 94. Consider the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f is differentiable on $\mathbb{R}$, but f' is not continuous at 0. Indeed,

$$f'(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h)}{h} = \lim_{h \rightarrow 0} h \sin(1/h) = 0.$$

For $x \neq 0$,

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

Since $\cos(1/x)$ oscillates as $x \rightarrow 0$, the derivative has no limit at 0. Hence f' is not continuous at 0.

Example 95. Consider

$$f(x) = \begin{cases} x + x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f is differentiable on $\mathbb{R}$ with

$$f'(0) = 1,$$

but

$$f'(x) = 1 + 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), \quad x \neq 0.$$

Again, f' oscillates near 0 and hence is not continuous there.

Theorem 44. (Algebra of Differentiation). Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable at $c \in (a, b)$, and let $\lambda \in \mathbb{R}$. Then:

(a) **(Linearity)**

$$(f + g)'(c) = f'(c) + g'(c), \quad (\lambda f)'(c) = \lambda f'(c).$$

(b) **(Product rule)**

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

(c) **(Quotient rule)** If $g(c) \neq 0$, then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Theorem 45 (Linear Approximation Theorem). If f is defined on (a, b) and differentiable at a point $c \in (a, b)$, then there is a function f^* (depending on f and on c) which is continuous at c and which satisfies

$$f(x) - f(c) = (x - c)f^*(x), \tag{1}$$

for all $x \in (a, b)$, with $f^*(c) = f'(c)$. Conversely, if there is a function f^* , continuous

at c , which satisfies (1) for all $x \in (a, b)$, then f is differentiable at c and $f'(c) = f^*(c)$.

Proof. If $f'(c)$ exists, define f^* on (a, b) by

$$f^*(x) = \begin{cases} \frac{f(x) - f(c)}{x - c}, & x \neq c, \\ f'(c), & x = c. \end{cases}$$

Then f^* is continuous at c and (1) holds for all $x \in (a, b)$.

Conversely, if (1) holds for some f^* continuous at c , then dividing by $x - c$ and letting $x \rightarrow c$ shows that $f'(c)$ exists and equals $f^*(c)$. $\square$

Example 96. Use the Linear Approximation Theorem to estimate $\sqrt{4.1}$.

Solution. Let $f(x) = \sqrt{x}$. Since f is differentiable at 4, the Linear Approximation Theorem gives

$$f(x) - f(4) = (x - 4)f^*(x),$$

where f^* is continuous at 4 and $f^*(4) = f'(4)$.

Since f^* is continuous, $f^*(x) \approx f^*(4) = f'(4)$ whenever x is close to 4. Thus,

$$f(x) \approx f(4) + f'(4)(x - 4).$$

Now, $f(4) = 2$ and $f'(4) = \frac{1}{4}$. Hence,

$$\sqrt{x} \approx 2 + \frac{1}{4}(x - 4)$$

for x near 4. Taking $x = 4.1$, we obtain

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025.$$

Theorem 46 (Chain Rule). Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable at $c \in (a, b)$, and let g be defined on (p, q) , where $f(c) \in (p, q)$ and g is differentiable at the point $f(c)$. Then the composition $g \circ f$ is differentiable at c , and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Proof. Since f is differentiable at c , by Theorem 45 there exists a function f^* , continuous at c , such that

$$f(x) - f(c) = (x - c)f^*(x),$$

for all $x \in (a, b)$, with $f^*(c) = f'(c)$.

Similarly, since g is differentiable at $f(c)$, there exists a function g^* , continuous at

$f(c)$, such that

$$g(y) - g(f(c)) = (y - f(c))g^*(y),$$

for all y near $f(c)$, with $g^*(f(c)) = g'(f(c))$.

Now substitute $y = f(x)$ to obtain

$$g(f(x)) - g(f(c)) = (f(x) - f(c))g^*(f(x)).$$

Using the expansion of $f(x) - f(c)$, we have

$$g(f(x)) - g(f(c)) = (x - c)f^*(x)g^*(f(x)).$$

Define

$$h^*(x) = f^*(x)g^*(f(x)).$$

Then

$$g(f(x)) - g(f(c)) = (x - c)h^*(x).$$

Since f^* is continuous at c and g^* is continuous at $f(c)$, it follows that h^* is continuous at c . Moreover,

$$h^*(c) = f^*(c)g^*(f(c)) = f'(c)g'(f(c)).$$

Therefore, by Theorem 45, $g \circ f$ is differentiable at c , and

$$(g \circ f)'(c) = h^*(c) = g'(f(c))f'(c).$$

□

Example 97. (a) Let $f(x) = x^2$ and $g(y) = \sin y$. Then

$$(g \circ f)(x) = \sin(x^2).$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \cos(x^2) \cdot (2x) = 2x \cos(x^2).$$

(b) Let $f(x) = e^x$ and $g(y) = \ln y$, defined on $(0, \infty)$. Then

$$(g \circ f)(x) = \ln(e^x) = x.$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \frac{1}{e^x} \cdot e^x = 1,$$

which matches the fact that the derivative of x is 1.

(c) Let $f(x) = \sqrt{x}$ on $(0, \infty)$ and $g(y) = y^3$. Then

$$(g \circ f)(x) = (\sqrt{x})^3 = x^{3/2}.$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = 3(\sqrt{x})^2 \cdot \frac{1}{2\sqrt{x}} = \frac{3}{2}x^{1/2},$$

which is indeed the derivative of $x^{3/2}$.

Exercise 22. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an even function, that is,

$$f(-x) = f(x) \quad \text{for all } x \in \mathbb{R}.$$

Assume that f is differentiable. Show that f' is an odd function.

Hint. Differentiate the identity $f(-x) = f(x)$ using the Chain Rule.

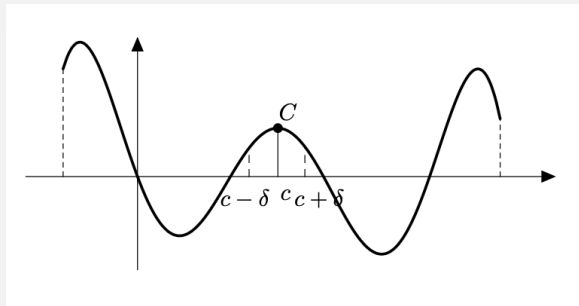
5.2 Mean value theorem

Definition 39 (Local Maximum and Local Minimum). Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \rightarrow \mathbb{R}$ be a function. A point $c \in J$ is called a local maximum of f if there exists $\delta > 0$ such that

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \leq f(c) \quad \text{for all } x \in (c - \delta, c + \delta).$$



Similarly, $c \in J$ is called a local minimum of f if there exists $\delta > 0$ such that

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \geq f(c) \quad \text{for all } x \in (c - \delta, c + \delta).$$

A point c is called a local extremum if it is either a local maximum or a local minimum.

Definition 40 (Global Maximum and Global Minimum). Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \rightarrow \mathbb{R}$ be a function.

A point $x_0 \in J$ is called a global maximum of f on J if

$$f(x) \leq f(x_0) \quad \text{for all } x \in J.$$

Similarly, $x_0 \in J$ is called a global minimum of f on J if

$$f(x) \geq f(x_0) \quad \text{for all } x \in J.$$

Do you believe that there is any relation between a global maximum and a local maximum? For example, is any global maximum necessarily a local maximum? Or is any local maximum necessarily a global maximum?

Example 98. Consider the following functions on $\mathbb{R}$:

1. **Local maximum.** Let $f(x) = -x^2$. At $c = 0$, we have $f(0) = 0$. For x close to 0, $f(x) = -x^2 \leq 0 = f(0)$. Hence, $c = 0$ is a local maximum point.
2. **Local minimum.** Let $f(x) = x^2$. At $c = 0$, we have $f(0) = 0$. For x close to 0, $f(x) = x^2 \geq 0 = f(0)$. Hence, $c = 0$ is a local minimum point.
3. **Saddle point.** Let $f(x) = x^3$. At $c = 0$, we have $f(0) = 0$. But in every neighborhood of 0, there are points $x > 0$ where $f(x) > 0 = f(0)$ and points $x < 0$ where $f(x) < 0 = f(0)$. Also $f'(c) = 0$. So $c = 0$ is neither a local maximum nor a local minimum; it is a saddle point.

Example 99. Consider the function $f : [a, b] \rightarrow \mathbb{R}$ defined by $f(x) = x$. Since $f(x) \leq b = f(b)$ for every $x \in [a, b]$, the point b is a global maximum of f . However, b is not a local maximum, as it is an endpoint of the interval. Similarly, a is a global minimum of f , but it is not a local minimum.

Exercise 23. Let $g : [-2\pi, 2\pi] \rightarrow \mathbb{R}$ be defined by $g(x) = \cos x$. Show that $x = 0$ is both a local maximum and a global maximum of g . What can you say about the points $x = \pm 2\pi$?

Theorem 47. [Fermat's Theorem] Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \rightarrow \mathbb{R}$ be differentiable at $c \in J$. If c is a local extremum of f , then

$$f'(c) = 0.$$

Proof. Assume that c is a local maximum of f . Then, for h sufficiently small,

$$f(c + h) \leq f(c).$$

Hence,

$$\frac{f(c + h) - f(c)}{h} \leq 0 \quad \text{if } h > 0,$$

and

$$\frac{f(c + h) - f(c)}{h} \geq 0 \quad \text{if } h < 0.$$

Passing to the limit as $h \rightarrow 0^+$ and $h \rightarrow 0^-$, we obtain

$$f'(c) \leq 0 \quad \text{and} \quad f'(c) \geq 0.$$

Therefore,

$$f'(c) = 0.$$

The proof for a local minimum is analogous. □

Example 100. *The assumption that c is an interior point is essential in Fermat's Theorem. If c lies on the boundary of the domain, the conclusion $f'(c) = 0$ may fail. For instance, let*

$$f : [0, \infty) \rightarrow \mathbb{R}, \quad f(x) = x.$$

At $c = 0$, we have $f(0) = 0$, and clearly

$$f(x) \geq f(0) \quad \text{for all } x \in [0, \infty),$$

so f has a local minimum at 0. However, $c = 0$ is not an interior point of the domain. Moreover, $f'(0) = 1 \neq 0$, which shows that the conclusion of Fermat's Theorem does not hold in this case.

Theorem 48. (Cauchy's Mean Value Theorem). *Let f and g are continuous real valued functions on $[a, b]$ which are differentiable in (a, b) , then there is a point $x \in (a, b)$ at which*

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x)$$

Proof. Take

$$h(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t), \quad (a \leq t \leq b).$$

Then h is continuous on $[a, b]$, h is differentiable in (a, b) , and

$$h(a) = f(b)g(a) - f(a)g(b) = h(b).$$

To prove the theorem, we have to show that $h'(x) = 0$ for some $x \in (a, b)$. If h is constant, this holds for every $x \in (a, b)$. If $h(t) > h(a)$ for some $t \in (a, b)$, let x be

a point on (a, b) at which h attains its global maximum by using the Extreme Value Theorem 35. Moreover, as $h(a) = h(b)$, so x must not be an endpoint, and hence x is a local maximum. Now, as $x \in (a, b)$, so applying Fermat's Theorem 47, we have $h'(x) = 0$.

If $h(t) < h(a)$ for some $t \in (a, b)$, we can apply the same argument where h attains a global minimum. $\square$

5.3 Applications

Suppose a function starts and ends at the same height. Do you expect its slope to be zero at some point in between? The answer is affirmative. This remarkable result is known as *Rolle's Theorem*. The theorem is named after the French mathematician Michel Rolle, who proved the result for polynomials in 1691 without using differential calculus. A rigorous proof in the modern setting was later given by Augustin-Louis Cauchy in 1823.

Theorem 49 (Rolle's Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a function satisfying:*

1. *f is continuous on $[a, b]$;*
2. *f is differentiable on (a, b) ;*
3. *$f(a) = f(b)$.*

Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

Proof. Apply Cauchy's Mean Value Theorem 48 with $g(x) = x$. Since g is continuous on $[a, b]$, differentiable on (a, b) , and $g'(x) = 1$, there exists $c \in (a, b)$ such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Since $f(a) = f(b)$, we obtain

$$0 = (b - a)f'(c).$$

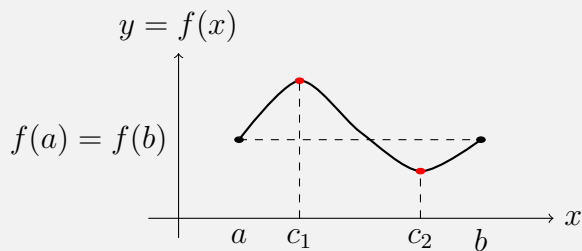
As $a < b$, we have $b - a > 0$. Hence,

$$f'(c) = 0.$$

This completes the proof. $\square$

Geometrically, the theorem guarantees the existence of a point where the tangent to the curve $y = f(x)$ is horizontal. Physically, if a particle returns to its initial position,

then at some intermediate time its instantaneous velocity must be zero.



Example 101. *The hypotheses of Rolle's Theorem are sufficient but not necessary. That is, a function may satisfy the conclusion of the theorem even if one or more of its hypotheses fail. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$. On the interval $[-1, 1]$, we have $f(-1) \neq f(1)$, so the hypothesis $f(a) = f(b)$ of Rolle's Theorem is not satisfied. Nevertheless,*

$$f'(x) = 3x^2,$$

and hence $f'(0) = 0$.

Thus, the conclusion of Rolle's Theorem holds even though one of its hypotheses fails.

Exercise 24. *Show that the continuity hypothesis in Rolle's Theorem cannot be omitted.*

Hint. Consider

$$f(x) = \begin{cases} 1, & x = -1, \\ x, & -1 < x \leq 1. \end{cases}$$

Verify that $f(-1) = f(1) = 1$ and that f is differentiable on $(-1, 1)$. Show that there is no $c \in (-1, 1)$ such that $f'(c) = 0$.

Exercise 25. *Show that the differentiability hypothesis in Rolle's Theorem cannot be omitted.*

Hint. Consider

$$f(x) = |x|, \quad -1 \leq x \leq 1.$$

Verify that $f(-1) = f(1)$ and that f is continuous on $[-1, 1]$. Show that there is no $c \in (-1, 1)$ such that $f'(c) = 0$.

Exercise 26. *Show that the condition $f(a) = f(b)$ in Rolle's Theorem cannot be omitted.*

Hint. Consider

$$f(x) = x, \quad -1 \leq x \leq 1.$$

Verify that f is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. Show that there is no $c \in (-1, 1)$ such that $f'(c) = 0$.

Theorem 50 (Lagrange's Mean Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a function satisfying:*

1. f is continuous on $[a, b]$;
2. f is differentiable on (a, b) .

Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Apply Cauchy's Mean Value Theorem 48 with $g(x) = x$. Since g is continuous on $[a, b]$, differentiable on (a, b) , and $g'(x) = 1$, there exists $c \in (a, b)$ such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

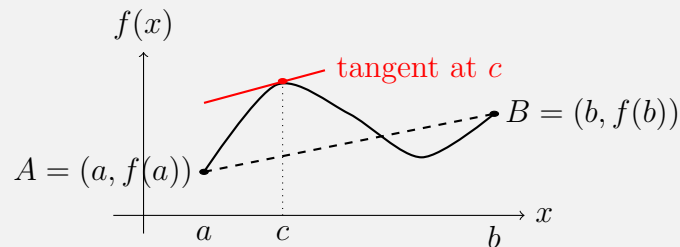
Since $g'(c) = 1$ and $g(b) - g(a) = b - a$, we obtain

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This completes the proof. □

Geometrically, it means that the tangent to the curve at $x = c$ is parallel to the secant line joining $(a, f(a))$ and $(b, f(b))$, that is,

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Example 102. *Show that*

$$|\sin x - \sin y| \leq |x - y|$$

for all $x, y \in \mathbb{R}$.

Solution. Assume $x \neq y$. By the Mean Value Theorem, there exists c between x and y such that

$$\sin x - \sin y = \cos(c)(x - y).$$

Taking absolute values,

$$|\sin x - \sin y| = |\cos(c)| |x - y| \leq |x - y|.$$

Example 103. *Show that for every $x > 0$,*

$$\ln(1 + x) < x.$$

Solution. Apply the Mean Value Theorem to $f(t) = \ln t$ on $[1, 1 + x]$. Then there exists $c \in (1, 1 + x)$ such that

$$\ln(1 + x) - \ln 1 = \frac{1}{c} x.$$

Since $c > 1$, we have $\frac{1}{c} < 1$. Hence

$$\ln(1 + x) = \frac{x}{c} < x.$$

Example 104. *Show that*

$$|e^x - e^y| \leq e^{\max\{x, y\}} |x - y|$$

for all $x, y \in \mathbb{R}$.

Solution. By the Mean Value Theorem, there exists c between x and y such that

$$e^x - e^y = e^c(x - y).$$

Since $e^c \leq e^{\max\{x, y\}}$, the result follows.

Example 105. *Show that the equation*

$$x^3 + x + 1 = 0$$

has at most one real solution.

Solution. Suppose $a \neq b$ are two solutions. By Rolle's Theorem, there exists $c \in (a, b)$ such that

$$3c^2 + 1 = 0,$$

which is impossible. Hence, the equation has at most one real solution.

Example 106. Suppose $f : J \rightarrow \mathbb{R}$ is differentiable on an interval J and

$$f'(x) = 0 \quad \text{for all } x \in J.$$

Show that f is constant on J .

Solution. Let $x, y \in J$ with $x < y$. Since J is an interval, $[x, y] \subseteq J$. By the Mean Value Theorem, there exists $c \in (x, y)$ such that

$$f(y) - f(x) = f'(c)(y - x).$$

Since $f'(c) = 0$, it follows that

$$f(y) = f(x).$$

As x and y are arbitrary, f is constant on J .

The assumption that J is an interval is essential in Example 106. For example, let

$$U = (-1, 0) \cup (0, 1),$$

and define

$$f(x) = \begin{cases} -1, & x \in (-1, 0), \\ 1, & x \in (0, 1). \end{cases}$$

Then f is differentiable on U and $f'(x) = 0$ for every $x \in U$, but f is not constant on U .

Example 107. Suppose $f : J \rightarrow \mathbb{R}$ is differentiable on an interval J .

1. If $f'(x) > 0$ for all $x \in J$, then f is strictly increasing on J .

Indeed, if $x < y$, then by the Mean Value Theorem there exists $c \in (x, y)$ such that $f(y) - f(x) = f'(c)(y - x)$. Since $f'(c) > 0$ and $y - x > 0$, we obtain $f(y) > f(x)$.

2. Similarly, if $f'(x) < 0$ for all $x \in J$, then f is strictly decreasing on J .

Example 108. Let $f : [2, 5] \rightarrow \mathbb{R}$ be continuous and differentiable on $(2, 5)$. Assume that

$$f'(x) = (f(x))^2 + \pi \quad \text{for all } x \in (2, 5).$$

True or false: $f(5) - f(2) = 3$.

Solution. The statement is false. By the Mean Value Theorem, there exists $c \in (2, 5)$ such that

$$f(5) - f(2) = f'(c)(5 - 2) = 3f'(c).$$

Since

$$f'(c) = (f(c))^2 + \pi > \pi,$$

we obtain

$$f(5) - f(2) > 3\pi > 3.$$

Hence, the given statement is false.

Example 109. Let $f : (0, 1] \rightarrow \mathbb{R}$ be differentiable and suppose that

$$|f'(x)| < 1 \quad \text{for all } x \in (0, 1].$$

Define

$$a_n = f\left(\frac{1}{n}\right).$$

Show that the sequence (a_n) converges.

Solution. By the Mean Value Theorem, for each $n \geq 1$, there exists $c_n \in (\frac{1}{n+1}, \frac{1}{n})$ such that

$$a_{n+1} - a_n = f\left(\frac{1}{n+1}\right) - f\left(\frac{1}{n}\right) = f'(c_n) \left(\frac{1}{n+1} - \frac{1}{n}\right).$$

Hence,

$$|a_{n+1} - a_n| \leq \left| \frac{1}{n+1} - \frac{1}{n} \right| = \frac{1}{n(n+1)}.$$

Therefore, for $m > n$,

$$|a_m - a_n| \leq \sum_{k=n}^{m-1} |a_{k+1} - a_k| \leq \sum_{k=n}^{m-1} \frac{1}{k(k+1)} = \frac{1}{n} - \frac{1}{m} \leq \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$, the sequence (a_n) is Cauchy in $\mathbb{R}$ and so by Theorem 10 it converges.

Example 110. Suppose that $f, g : J \rightarrow \mathbb{R}$ are differentiable on an interval J and satisfy

$$f'(x) = g'(x), \quad \forall x \in J.$$

Then there exists a constant $C \in \mathbb{R}$ such that

$$f(x) = g(x) + C, \quad \forall x \in J.$$

Solution. Define

$$h(x) = f(x) - g(x).$$

Then

$$h'(x) = f'(x) - g'(x) = 0, \quad \forall x \in J.$$

By the previous result, h is constant on J . Hence there exists $C \in \mathbb{R}$ such that

$$h(x) = C,$$

that is,

$$f(x) = g(x) + C, \quad \forall x \in J.$$

Example 111. Suppose that $f : J \rightarrow \mathbb{R}$ is differentiable on an interval J and satisfies

$$0 < m \leq f'(x) \leq M, \quad \forall x \in J,$$

where $m, M > 0$. Show that for every $x, y \in J$ with $x < y$,

$$m(y - x) \leq f(y) - f(x) \leq M(y - x).$$

Solution. By the Mean Value Theorem, there exists $c \in (x, y)$ such that

$$f(y) - f(x) = f'(c)(y - x).$$

Since

$$m \leq f'(c) \leq M$$

and

$$y - x > 0,$$

we obtain

$$m(y - x) \leq f(y) - f(x) \leq M(y - x).$$

Example 112. Suppose that $f : J \rightarrow \mathbb{R}$ is differentiable on an interval J and satisfies

$$f'(x) > 0, \quad \forall x \in J.$$

Show that f is one-to-one.

Solution. From the previous example, f is strictly increasing on J . Hence, whenever $x_1 < x_2$,

$$f(x_1) < f(x_2).$$

Therefore,

$$f(x_1) = f(x_2)$$

implies

$$x_1 = x_2.$$

Thus f is one-to-one.

Exercise 27. Suppose f is a real differentiable function on $[a, b]$ and suppose

$$f'(a) < \lambda < f'(b).$$

Then there is a point $x \in (a, b)$ such that $f'(x) = \lambda$. A similar result holds, of course, if $f'(a) > f'(b)$. This is known as (**Darboux's theorem**).

Sketch of the proof.

- Set $g(t) = f(t) - \lambda t$.
- Then $g'(a) < 0$, so that $g(t_1) < g(a)$ for some $t_1 \in (a, b)$, and $g'(b) > 0$, so that $g(t_2) < g(b)$ for some $t_2 \in (a, b)$.
- Use EVT 35 and Fermat's Theorem 47.

Exercise 28. Suppose f is differentiable on $[0, 2]$ with $f'(0) = 7$ and $f'(2) = 3$. For what values of λ can you guarantee the existence of some $c \in (0, 2)$ such that

$$f'(c) = \lambda?$$

Derivatives of Higher Order. Higher-order derivatives provide deeper insights into the behavior of functions beyond what the first derivative reveals. Geometrically, while the first derivative measures slope and the second captures curvature, higher derivatives describe subtler features of the graph. They are crucial for classifying critical points when lower-order tests fail, and they naturally arise in applications as successive rates of change—velocity, acceleration, jerk, and beyond. Moreover, higher derivatives underpin approximation methods like Taylor series and play an essential role in advanced areas of mathematics, including differential equations, geometry, and mathematical physics, making them a fundamental tool for both theory and applications.

Definition 41. Let $f : (a, b) \rightarrow \mathbb{R}$ be a function. If f is differentiable on (a, b) and f' is also differentiable, then we call the derivative of f' the **second derivative** of f , denoted by f'' . Inductively, if $f^{(n-1)}$ exists and is differentiable, then its derivative is called the n -th derivative of f , denoted by $f^{(n)}$.

Example 113. It is not always true that all higher-order derivatives exist. The function $f(x) = |x|^3$ is differentiable everywhere and twice differentiable everywhere, but $f'''(0)$ does not exist. The function $f(x) = e^x$ has derivatives of all orders, given by $f^{(n)}(x) = e^x$. This type of function is called a smooth function.

Theorem 51. Let $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable at $c \in (a, b)$.

1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .
3. If $f'(c) = 0$ and $f''(c) = 0$, the test is inconclusive.

Proof. Assume $f'(c) = 0$. Then by the definition of the second derivative,

$$f''(c) = \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c)}{h} = \lim_{h \rightarrow 0} \frac{f'(c+h)}{h}.$$

Case 1: $f''(c) > 0$. Then for h small and positive, $\frac{f'(c+h)}{h} > 0$. This means:

- If $h > 0$, then $f'(c+h) > 0$, so f is increasing to the right of c .
- If $h < 0$, then $f'(c+h) < 0$, so f is decreasing to the left of c .

Thus f decreases before c and increases after c , giving a local minimum.

Case 2: $f''(c) < 0$. By the same reasoning, $f'(c+h) < 0$ for $h > 0$ and $f'(c+h) > 0$ for $h < 0$, so f increases before c and decreases after c , giving a local maximum.

Case 3: $f''(c) = 0$. Then the sign argument above gives no definite change of monotonicity, so no conclusion can be drawn. $\square$

Example 114. • $f(x) = x^2$: $f'(0) = 0$, $f''(0) = 2 > 0 \implies$ local minimum at 0.

- $f(x) = -x^2$: $f'(0) = 0$, $f''(0) = -2 < 0 \implies$ local maximum at 0.
- $f(x) = x^3$: $f'(0) = 0$, $f''(0) = 0 \implies$ inconclusive (in fact, stationary inflection point).

Taylor's Theorem. From optimizing machine learning algorithms to improving the precision of space navigation, the Taylor series is fundamental to advancing modern technology. Additionally, it aids in numerical solutions for differential equations, computational fluid dynamics, and medical imaging.

Theorem 52. Suppose f is a real function on $[a, b]$, n is a positive integer, $f^{(n-1)}$ is continuous on $[a, b]$, $f^{(n)}(t)$ exists for every $t \in (a, b)$. Let α, β be distinct points of (a, b) , and define

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k$$

Then there exists a point x between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$$

Proof. Let M be the number defined by

$$f(\beta) = P(\beta) + M(\beta - \alpha)^n$$

and put

$$g(t) = f(t) - P(t) - M(t - \alpha)^n \quad (a \leq t \leq b)$$

We have to show that $n!M = f^{(n)}(x)$ for some x between α and β . By the definition of $P(t)$ and above,

$$g^{(n)}(t) = f^{(n)}(t) - n!M \quad (a < t < b)$$

Hence, the proof will be complete if we can show that $g^{(n)}(x) = 0$ for some x between α and β .

Since $P^{(k)}(\alpha) = f^{(k)}(\alpha)$ for $k = 0, \dots, n-1$, we have

$$g(\alpha) = g'(\alpha) = \dots = g^{(n-1)}(\alpha) = 0.$$

Our choice of M shows that $g(\beta) = 0$, so that $g'(x_1) = 0$ for some x_1 between α and β , by the Mean value theorem. Since $g'(x_1) = 0$, we conclude similarly that $g''(x_2) = 0$ for some x_2 between α and x_1 . After n steps we arrive at the conclusion that $g^{(n)}(x_n) = 0$ for some x_n between α and x_{n-1} , that is, between α and β . $\square$

For $n = 1$, this is just the mean value theorem. In general, the theorem shows that f can be approximated by a polynomial of degree $n-1$, and the series expression allows us to estimate the error if we know bounds on $|f^{(n)}(x)|$

Thanks!